# Matrix-product bound for witness DAGs in commutative stochastic local search

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Editorial note. Original mathematical body, witness-DAG vector figure and bibliography retained. Only selected complete-proof matrix-witness lemma counted. Direct runtime/standard criterion consequences and explicitly author-omitted application proofs not counted. Source version includes publisher recorded2025-11-23 postpublication correction; no collector proof repair.

Allowed standard prerequisites (editorial).

- Discrete probability, transition matrices and directed acyclic graphs. Native flaw-selection algorithm, dependence relation, matrix commutativity, witness-DAG definitions and complete local proof are retained.. History-dependent conditional recursion and matrix argument remain local obligations, not editorial background.

## 1 Introduction

The Lovász Local Lemma (LLL) is a powerful tool in probabilistic combinatorics [9]. At a high level, it states that given a collection of bad events in a probability space $\mu$, where each bad event is not too likely and is independent of most other bad events, there is a strictly positive probability of avoiding all of them. In particular, a configuration avoiding all such bad events exists.

In its simplest, "symmetric" form, the LLL requires that each bad event have probability at most $p$ and is dependent with at most $d$ others, where $epd \leq 1$.

For example, consider a CNF formula where each clause has $k$ literals and shares variables with at most $L$ other clauses. For each clause $c$ we

can define the bad event $B_c$ that $c$ is violated in a chosen assignment of the variables. For a uniformly random variable assignment, each bad event has probability $p = 2^{-k}$ and affects at most $d \leq L$ others. So when $L \leq 2^k/\mathrm{e}$, the formula is satisfiable; crucially, this criterion does not depend on the total number of variables.

A generalization known as the Lopsided LLL (LLLL) allows bad events to be positively correlated (in a certain technical sense) instead of independent. For example, consider an $n \times n$ array of colors, where each color appears at most $\Delta$ times in total. We wish to find a latin transversal of $C$, that is, a permutation $\pi$ over $\{1, \ldots, n\}$ such that all colors $C(i, \pi i) : i = 1, \ldots, n$ are distinct. To apply the LLLL, we take our probability space to be the uniform distribution on permutations $\pi$. For each pair of cells $(x_1, y_1), (x_2, y_2)$ of the same color, there is a corresponding bad event $\pi x_1 = y_1 \wedge \pi x_2 = y_2$, which has probability $p = \frac{1}{n(n-1)}$. Erdős & Spencer [10] showed that two bad events here are negatively correlated only if they overlap on a row or column. So each bad event is dependent with at most $d = 4(\Delta - 1)n$ others. Thus, for $\Delta \leq \frac{n}{4\mathrm{e}}$, the LLLL criterion holds and a transversal exists. A stronger form of the LLLL (the cluster-expansion criterion [5]) tightens this to $\Delta \leq \frac{27n}{256}$, which is the strongest bound currently known.

Although the LLLL applies to general probability spaces, most constructions in combinatorics use a simpler setting we refer to as the variable LLL. Here, the probability space $\mu$ is a Cartesian product with $n$ independent variables $X = (X_1, \ldots, X_n)$, and each bad event is determined by a subset of the variables. Two bad events are dependent if they share a common variable. This covers, for instance, the CNF formula example described above. In a seminal paper, Moser & Tardos [26] presented a simple local search algorithm to make the variable LLL constructive. This algorithm can be described as follows.

Algorithm 1 The Moser-Tardos (MT) resampling algorithm

```
Draw the state X from distribution μ
while some bad event is true on X do
    Select, arbitrarily, a bad event B true on X
    Resample, according to distribution μ, all variables X_i affecting B
```

Moser & Tardos showed that if the symmetric LLL criterion (or the more general asymmetric LLLL criterion) is satisfied, then this algorithm quickly converges. Following this result, a large effort has been devoted to making different variants of the LLLL constructive. This research has taken many

directions, including further analysis of Algorithm 1 and its connection to different LLL criteria [6, 23, 27].

One line of research has been to use variants of the MT algorithm for general probability spaces beyond the variable LLL. We can summarize this in the following framework. There is a discrete state space $\Omega$, with a collection $\mathcal{F}$ of subsets of $\Omega$ which we call flaws. There is also a randomized procedure called the resampling oracle $\mathrm{R}_f$ for each flaw $f$; it takes some random action to attempt to "fix" that flaw, resulting in a new state $\sigma' \leftarrow \mathrm{R}_f(\sigma)$. (It is possible that this new state $\sigma'$ does not actually fix $f$.) With these ingredients, we define the general local Search Algorithm as follows.

Algorithm 2 The Search Algorithm

1: Draw the state $\sigma$ from some distribution $\mu$
2: while some flaw holds on $\sigma$ do
3: Select a flaw $f$ of $\sigma$, according to some rule S.
4: Update $\sigma \leftarrow \mathrm{R}_f(\sigma)$.

Again consider the latin transversal construction, which was one of the main original motivations behind the Search Algorithm [18]. For this problem, each pair of cells with the same color now corresponds to a flaw. The resampling oracle for a flaw applies a random "swapping" operation to the permutation. This algorithm generates a latin transversal under the same conditions as the existential LLLL argument. Further application of the Search Algorithm include matchings and spanning trees of the clique [1, 2, 18, 22, 20] as well as settings not directly connected to the LLL [3, 7, 19].

The most important question about the behavior of the Search Algorithm is whether it converges to a flawless object. But, there are other important questions to ask; for instance, "is it parallelizable?", "how many solutions can it output?", "what is the expected 'weight' of a solution?". These questions and more have been answered for the MT algorithm in a long series of papers [6, 8, 11, 12, 15, 17, 23, 26]. For example, results of [12, 14, 17] discuss how to estimate the entropy of the output distribution and how to deal algorithmically with super-polynomially many bad events.

We emphasize the role of the selection rule S in the Search Algorithm. This procedure must choose which flaw $f \ni \sigma$ to resample, if there are multiple possibilities; it may depend on the prior states and may be randomized. The original MT algorithm allows nearly complete freedom for this. However, for general resampling oracles, convergence guarantees are only known

for a few relatively rigid rules such as selecting the flaw of least index [20]. In [24], Kolmogorov identified a property called commutativity that allows a free choice for S in the Search Algorithm (as in the MT algorithm). This seemingly minor detail turns out to play a key role in extending the additional properties of the MT algorithm to the Search Algorithm. For instance, it leads to parallel algorithms [24] and to bounds on the output distribution [21].

At a high level, the goal of this paper is to provide a more conceptual, algebraic explanation for the commutativity properties of resampling oracles and their role in the Search Algorithm. We do this by introducing a notion of commutativity, essentially matrix commutativity, that is both more general and simpler than the definition in [24]. Most of our results had already been shown, in slightly weaker forms, in prior work [24, 21, 14]. However, the proofs were computationally heavy and narrowly targeted to certain probability spaces, with numerous technical side conditions and restrictions.

Before the formal definitions, let us provide some intuition. For each flaw $f$, consider an $|\Omega| \times |\Omega|$ transition matrix $A_f$. Each row of $A_f$ describes the probability distribution of resampling $f$ at a given state $\sigma$. We call the resampling oracle commutative if the transition matrices commute for each pair of flaws which are "independent" in the LLL sense. We show a number of results for such commutative oracles.

1. We obtain bounds on the distribution of the state at the termination of the Search Algorithm. These bounds are comparable to the LLL-distribution, i. e., the distribution induced by conditioning on avoiding all bad events. Similar results, albeit with a number of additional technical restrictions, had been shown in [21] for the original definition of commutativity.

2. We develop a generic parallel version of the Search Algorithm. This extends results of [24, 15], with simpler and more general proofs.

3. In many settings, flaws are formed from smaller "atomic" events [15]. We show that, if the atomic events satisfy the generalized commutativity definition, then so do the larger "composed" events. This natural property did not seem to hold for the original commutativity definition of [24].

4. For some probability spaces, specialized distributional bounds are available, beyond the "generic" LLL bounds [14]. Our construction captures many of these results in a stronger and more unified way.

As a concrete example of our results, we show that, for the latin transversal application, the resulting permutation $\pi$ has nice distributional properties. In particular, we show the following.

Theorem 1.1. If each color appears at most $\Delta \leq \frac{27}{256}n$ times in the array, then the Search Algorithm generates a latin transversal $\pi$ such that, for every pair $(x, y)$, it holds that

$$0.53/n \leq \Pr(\pi x = y) \leq 1.59/n\,.$$

The upper bound improves quantitatively over a similar bound of [14]; to the best of our knowledge, no non-trivial lower bound was previously known. Intriguingly, these bounds are not known for the LLL-distribution itself. To better situate Theorem 1.1, note that Alon, Spencer, & Tetali [4] showed that there is a (minuscule) universal constant $\beta > 0$ such that, if each color appears at most $\Delta = \beta n$ times in the array and $n$ is a power of two, then the array can be partitioned into $n$ independent transversals. In this case, randomly selecting a transversal from this list would give $\Pr(\pi x = y) = 1/n$. Theorem 1.1 can be regarded as a simplified fractional analogue, i. e., we fractionally decompose the given array into transversals, with $\Pr(\pi x = y) = \Theta(1/n)$ for all pairs $x, y$. Furthermore, we achieve this guarantee automatically, merely by running the Search Algorithm.

## 1.1 Overview of our approach

Although it will require significant definitions and technical development to state our results formally, let us provide a high-level summary here. As a starting point, consider the MT algorithm. Moser & Tardos [26] used a construction called a witness tree for the analysis: for each resampling of a bad event $B$, they generate a corresponding witness tree which records an "explanation" of why $B$ was true at that time. More properly, it provides a history of all prior resamplings which affected the variables involved in $B$.

The main technical lemma governing the behavior of the MT algorithm is the "Witness Tree Lemma," which states that the probability of producing a given witness tree is at most the product of the probabilities of the corresponding events. The bounds on algorithm running time, as well as parallel algorithms and distributional properties, then follow from a union bound over witness trees.

Versions of this Witness Tree Lemma have been shown for some variants of the MT algorithm [13, 19]. Iliopoulos [21] further showed that it held

for general spaces which satisfy the commutativity condition; this, in turn, leads to the nice algorithmic properties such as parallel algorithms.

Our main technical innovation is to generalize the Witness Tree Lemma. Instead of tracking a scalar product of probabilities in a witness tree, we consider a matrix product. We bound the probability of a given witness tree (or, more properly, a slight generalization known as a witness directed acyclic graph) in terms of the products of the transition matrices of the corresponding flaws. At the end, we obtain the scalar form of the Witness Tree Lemma by projecting to a one-dimensional eigenspace; for this, we take advantage of some spectral estimation methods of [3].

### 1.2 Outline of the paper

In Section 2, we introduce our new matrix-based definition for commutativity.

In Section 3, we define the witness directed acyclic graph following [11]. We show bounds on the Search Algorithm in terms of certain associated matrix products. We also discuss how these relate to standard "scalar" LLL criteria.

In Section 4, we derive simple bounds on the state distribution of the Search Algorithm.

In Section 5, we describe a parallel implementation of the Search Algorithm.

In Section 6, we consider a construction for building resampling oracles out of smaller "atomic" events.

In Section 7, we consider more involved distributional bounds, including applications to latin transversals and clique perfect matchings.

### 1.3 Definitions

Throughout the paper we consider implementations of the Search Algorithm. We list the resampled flaws in order as $f_0, f_1, f_2, \ldots$; to avoid ambiguity, we refer to the state at time $t$ as the state $\sigma$ just before resampling flaw $f_t$. Note that the state at time 0 is drawn directly from $\mu$.

For flaw $f$ and states $\sigma \in f, \sigma' \in \Omega$ we write $\sigma \xrightarrow{f} \sigma'$ to denote that the algorithm resamples $f$ at $\sigma$ and moves to $\sigma'$. We define $A_f[\sigma, \sigma']$ to be the probability of such transition under resampling oracle $\mathrm{R}_f$, i. e.,

$$A_f[\sigma, \sigma'] = \Pr(\mathrm{R}_f(\sigma) = \sigma') \,.$$

To allow us to write our state transitions from left-to-right, we also write $e_\sigma^\top A_f e_{\sigma'} = A_f[\sigma, \sigma']$. For $\sigma \notin f$, we define $A_f[\sigma, \sigma'] = 0$. Note that every vector $\theta$ over $\Omega$ satisfies $||\theta^\top A_f||_1 = \sum_{\sigma \in f} \theta[\sigma] \leq ||\theta^\top||_1$. Thus, the matrix $A_f$ is substochastic.

For an arbitrary event $E \subseteq \Omega$, we define $e_E$ to be the indicator vector for $E$, i. e., $e_E[\sigma] = 1$ if $\sigma \in E$ and $e_E[\sigma] = 0$ otherwise. For a state $\sigma \in \Omega$, we write $e_\sigma$ as shorthand for $e_{\{\sigma\}}$, i. e., the basis vector which has a 1 in position $\sigma$ and zero elsewhere. With this notation, $e_\sigma^\top A_f$ for $\sigma \in f$ is the vector representing the probability distribution obtained by resampling flaw $f$ at state $\sigma$. When $\sigma \notin f$, then $e_\sigma^\top A_f = \vec{0}$.

For vectors $u, v$ we write $u \preceq v$ if $u[i] \leq v[i]$ for all entries $i$. We write $u \propto v$ if there is some scalar value $c \geq 0$ with $u = cv$. Likewise, for matrices $A, B$ we write $A \propto B$ if $A = cB$ for some scalar value $c \geq 0$.

## 2 Commutativity

We suppose that we have fixed a symmetric relation $\sim$ on the set of flaws $\mathcal{F}$, with $f \not\sim f$ for all $f \in \mathcal{F}$. We refer to $\sim$ as the dependence relation. We define $\Gamma(f)$ to be the set of flaws $g$ with $f \sim g$, and we also define $\overline{\Gamma}(f) = \Gamma(f) \cup \{f\}$. We write $f \simeq g$ if $f \sim g$ or $f = g$.

A major contribution of this paper is the introduction of a natural definition for commutativity.

Definition 2.1 (Matrix commutativity). The resampling oracle is matrix-commutative with respect to dependence relation $\sim$ if $A_f A_g = A_g A_f$, for every pair of flaws $f, g$ such that $f \nsim g$.

For contrast, let us recall the definition of commutativity from [24]. To avoid confusion, we refer to this other notion as strong commutativity.

Definition 2.2 (Strong commutativity [24]). The resampling oracle is strongly commutative with respect to dependence relation $\sim$ if for every pair of flaws $f, g$ such that $f \nsim g$, there is an injective mapping from transitions $\sigma_1 \xrightarrow{f} \sigma_2 \xrightarrow{g} \sigma_3$ to transitions $\sigma_1 \xrightarrow{g} \sigma_2' \xrightarrow{f} \sigma_3$, so that $A_f[\sigma_1, \sigma_2] A_g[\sigma_2, \sigma_3] = A_g[\sigma_1, \sigma_2'] A_f[\sigma_2', \sigma_3]$.

Observation 2.3. If the resampling oracle is strongly commutative, then it is matrix-commutative.

Proof. Consider $f \not\sim g$. By symmetry, we need to show that $(A_f A_g)[\sigma, \sigma'] \leq (A_g A_f)[\sigma, \sigma']$ for any states $\sigma, \sigma'$. Let $V$ denote the set of states $\sigma''$ with

$A_f[\sigma,\sigma'']A_g[\sigma'',\sigma'] > 0$. By definition, there is an injective function $F : V \to \Omega$ such that $A_f[\sigma,\sigma'']A_g[\sigma'',\sigma'] = A_g[\sigma,F(\sigma'')]A_f[F(\sigma''),\sigma']$. Therefore, we have

$$(A_fA_g)[\sigma,\sigma'] = \sum_{\sigma''\in V} A_f[\sigma,\sigma'']A_g[\sigma'',\sigma'] = \sum_{\sigma''\in V} A_g[\sigma,F(\sigma'')]A_f[F(\sigma''),\sigma'].$$

Since $F$ is injective, each term $A_g[\sigma,\tau]A_f[\tau,\sigma']$ is counted at most once in this sum with $\tau = F(\sigma'')$. So $(A_fA_g)[\sigma,\sigma'] \le \sum_{\tau\in f} A_g[\sigma,\tau]A_f[\tau,\sigma'] = (A_gA_f)[\sigma,\sigma']$. □

Observation 2.4. Suppose the resampling oracle is matrix-commutative. If resampling flaw $f$ can cause flaw $g$, then $f \sim g$.

Proof. Consider some state transition $\sigma \xrightarrow{f} \sigma'$ for $\sigma \notin g, \sigma' \in g$. So $e_\sigma^\top A_f e_g > 0$. Then $e_\sigma^\top A_f A_g \neq \vec{0} = e_\sigma^\top A_g A_f$. So $A_fA_g \neq A_gA_f$. □

For the remainder of this paper, we use the word "commutative" to mean matrix-commutative. We always assume that the resampling oracle R is matrix-commutative unless explicitly stated otherwise.

We say that a set or multiset $I \subseteq \mathcal{F}$ is stable if $f \not\sim g$ for all distinct pairs $f, g \in I$. We define

$$A_I = \prod_{f\in I} A_f;$$

here, if $I$ is a multiset, each $f \in I$ is counted with its multiplicity in $I$. When the resampling oracle is commutative, this is well-defined (without specifying ordering of $I$) since the matrices $A_f$ all commute.

For a multiset $I = \{g_1, \dots, g_t\}$, we define $\langle I\rangle = g_1 \cap \cdots \cap g_t$.

For a flaw $f$, we write $f \sim I$ if there exists $g \in I$ with $f \sim g$. Similarly, for multisets $I, J$, we write $I \sim J$ if there exist $f \in I, g \in J$ with $f \sim g$.

# 3 Witness directed acyclic graphs and matrix bounds

Following [11], the witness directed acyclic graph is our key tool to analyze the Search Algorithm.

Definition 3.1 (Witness directed acyclic graph). A witness directed acyclic graph (abbreviated wdag) is a directed acyclic graph $H$, where each vertex $v \in H$ has a label $L(v)$ from $\mathcal{F}$, and such that for each pair of vertices $v, w \in H$, there is an edge between $v$ and $w$ (in either direction) if and only if $L(v) \simeq L(w)$.

For intuition, every node $v$ in a wdag corresponds to a resampled flaw $f$, with $L(v) = f$.

Although the definition of wdags does not specify a time ordering, in the wdags we consider the edges always point forward in time: there is an edge from $v$ to $w$ if the flaw corresponding to $v$ was resampled before $w$. For example, consider a scenario with five resampled flaws $f_1, f_2, f_3, f_1, f_5$ in order (where $f_4 = f_1$ has been resampled twice). We could represent this with the following wdag:

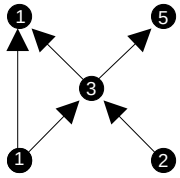



Figure 1: A wdag representing the five indicated flaws.

We define $\mathfrak{W}$ to be the set of wdags. The number of nodes in a wdag $H$ is denoted by $|H|$. Given a flaw $f$ and wdag $H$, we write $f \sim H$ if there exists a node $v \in H$ with $L(v) \sim f$.

For a wdag $H$ with sink nodes $v_1, \ldots, v_k$ labeled $f_1 = L(v_1), \ldots, f_k = L(v_k)$, observe that $f_1, \ldots, f_k$ are distinct and that $\{f_1, \ldots, f_k\}$ is a stable set of flaws; we denote it by $\text{sink}(H)$, and we define $\mathfrak{W}(I)$ to be the set of wdags with $\text{sink}(H) = I$. We also write $\mathfrak{W}(f)$ as shorthand for $\mathfrak{W}(\{f\})$. We define $\mathfrak{S} = \bigcup_{f \in \mathcal{F}} \mathfrak{W}(f)$ to be the set of single-sink wdags.

There is a key connection between wdags and transition matrices. For a wdag $H$, we define an associated $|\Omega| \times |\Omega|$ matrix $A_H$ inductively, as follows. If $H = \emptyset$, then $A_H$ is the identity matrix on $\Omega$. Otherwise, we choose an arbitrary source node $v$ of $H$ and set $A_H = A_{L(v)} A_{H-v}$.

Proposition 3.2. The definition of $A_H$ does not depend on the chosen source node $v$.

Proof. We show it by induction on $|H|$. When $|H| = 0$ it is vacuous. For the induction step, suppose $H$ has two source nodes $v_1, v_2$. We need to show that we get the same result by recursing on $v_1$ or $v_2$, i.e $A_{L(v_1)} A_{H-v_1} = A_{L(v_2)} A_{H-v_2}$.

We can apply the induction hypothesis to $H - v_1$ and $H - v_2$, noting that $v_2$ is a source node of $H - v_1$ and $v_1$ is a source node of $H - v_2$. We get $A_{H-v_1} = A_{L(v_2)} A_{H-v_1-v_2}, A_{H-v_2} = A_{L(v_1)} A_{H-v_1-v_2}$. Thus, in order to show $A_{L(v_1)} A_{H-v_1} = A_{L(v_2)} A_{H-v_2}$, it suffices to show that $A_{L(v_1)} A_{L(v_2)} =$

$A_{L(v_2)}A_{L(v_1)}$. Since $v_1, v_2$ are both source nodes, we have $L(v_1) \not\sim L(v_2)$. Thus, this follows from commutativity. □

Observation 3.3. If $f \not\sim H$ for a flaw $f$ and wdag $H$, then $A_f A_H = A_H A_f$.

Proof. We can write $A_H = A_{L(v_1)} \cdots A_{L(v_t)}$ where $v_1, \ldots, v_t$ are the nodes of $H$. By hypothesis, we have $L(v_i) \not\sim f$ for all $i$ and the matrices $A_{L(v_i)}$ all commute with $A_f$. So $A_f A_H = A_f A_{L(v_1)} \cdots A_{L(v_t)} = A_{L(v_1)} \cdots A_{L(v_t)} A_f = A_H A_f$. □

One important way of generating wdags is the following: we say that we prepend a given flaw $f$ to a given wdag $H$ when we add a node labeled $f$ with an edge to each other node $v$ with $L(v) \simeq f$. This operation always results in a new wdag $H'$, which has a source node labeled $f$. Intuitively, this means that $f$ happens before all the events corresponding to the nodes in $H$.

Observation 3.4. If wdag $H'$ is obtained by prepending flaw $f$ to wdag $H$, then $A_{H'} = A_f A_H$.

## 3.1 Wdags and their role in the Search Algorithm

Consider an execution of the Search Algorithm. For each time $t$ with resampled flaw $f_t$, we define a corresponding wdag $G_t$ as follows. Initially, we set $G_t$ to consist of a singleton node labeled $f_t$. Then, for each time $s = t-1, \ldots, 0$ with resampled flaw $f_s$, there are two cases:

- If $f_s \sim G_t$, or if $G_t$ has a source node labeled $f_s$, then prepend $f_s$ to $G_t$.
- Otherwise, do not modify $G_t$.

Note that $\mathrm{sink}(G_t) = \{f_t\}$. We say a wdag $H$ appears if $H$ is isomorphic to some $G_t$; with a slight abuse of notation, we write this simply as $G_t = H$.

This construction is very similar to the Witness Tree in Moser & Tardos [26]. One of the main ingredients in their proof is the observation that a witness tree $G$ appears with probability at most $\prod_{v \in G} \mu(L(v))$, i. e., the product of probabilities of the flaws that label its vertices. (Recall that $\mu$ denotes the initial state distribution.) Their proof depends on properties of the variable setting and does not extend to other probability spaces. Our key message is that the new commutativity definition allows us to analogously bound the probability of appearance of a given wdag by the product of transition matrices. Specifically, we show the following.

Lemma 3.5. For any wdag $H$, the probability that $H$ appears is at most $\mu^\top A_H \vec{1}$.

Proof. We first show that if the Search Algorithm runs for at most $t_{\max}$ steps starting with state $\sigma$, where $t_{\max}$ is an arbitrary integer, then $H$ appears with probability at most $e_\sigma^\top A_H \vec{1}$. We prove this claim by induction on $t_{\max}$. The claim holds vacuously for $t_{\max} = 0$, or if $\sigma$ is flawless. Also, if $H$ is a singleton node labeled $f$ and $\sigma \in f$, then $e_\sigma^\top A_H \vec{1} = 1$ and the claim is vacuous.

So suppose that $t_{\max} > 0$ and $\sigma$ has a flaw, and that $H$ is not a singleton node with label $f \ni \sigma$. Then $G_0 \neq H$. So, if $H$ appears, we have $G_t = H$ for $t \in \{1, \ldots, t_{\max} - 1\}$. Now suppose we condition on S selecting a flaw $f$ to resample in $\sigma$. We can view the evolution of the Search Algorithm A as a two-part process: first resample $f$, reaching a state $\sigma'$, wherein each potential choice of $\sigma'$ is chosen with probability $A_f[\sigma, \sigma']$. Then execute a new search algorithm A$'$ starting at state $\sigma'$, wherein the flaw selection rule S$'$ on history $(\sigma', \sigma_2, \ldots, \sigma_t)$ is the same as the choice of S on history $(\sigma, \sigma', \sigma_2, \ldots, \sigma_t)$.

Let $G'_{t-1}$ be the corresponding wdag for A$'$. We obtain $G_t$ from $G'_{t-1}$ either by prepending $f$, or by setting $G_t = G'_{t-1}$ when $f \not\sim G'_{t-1}$ and $G'_{t-1}$ has no source node labeled $f$. Consequently, $H$ must satisfy one of the following two mutually exclusive conditions: (i) $H$ has a unique source node $v$ labeled $f$ and $G'_{t-1} = H - v$; or (ii) $f \not\sim H$ and $G'_{t-1} = H$.

In case (i), if $H$ appears for the original search algorithm A within $t_{\max}$ timesteps, then $H - v$ must appear for A$'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most $e_{\sigma'}^\top A_{H-v} \vec{1}$ for a fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of

$$\sum_{\sigma'} A_f[\sigma, \sigma'] e_{\sigma'}^\top A_{H-v} \vec{1} = e_\sigma^\top A_f A_{H-v} \vec{1} = e_\sigma^\top A_H \vec{1}$$

as required.

In case (ii), note that by Observation 3.3 the matrices $A_f$ and $A_H$ commute. Again, if $H$ appears for the original search algorithm A within $t_{\max}$ timesteps, then $H$ must appear for A$'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most $e_{\sigma'}^\top A_H \vec{1}$ for a fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of

$$\sum_{\sigma'} A_f[\sigma, \sigma'] e_{\sigma'}^\top A_H \vec{1} = e_\sigma^\top A_f A_H \vec{1} = e_\sigma^\top A_H A_f \vec{1}.$$

Since $A_f$ is substochastic, this is at most $e_\sigma^\top A_H \vec{1}$.

This completes the induction. By countable additivity, we can compute the probability that $H$ ever appears from starting state $\sigma$, as

$$\Pr(\bigvee_{t=0}^{\infty} G_t = H) = \lim_{t_{\max}\to\infty} \Pr(\bigvee_{t=0}^{t_{\max}-1} G_t = H) \leq \lim_{t_{\max}\to\infty} e_\sigma^\top A_H \vec{1} = e_\sigma^\top A_H \vec{1}.$$

Finally, integrating over $\sigma$ gives $\sum_\sigma \mu[\sigma] e_\sigma^\top A_H \vec{1} = \mu^\top A_H \vec{1}$. □

Corollary 3.6. The expected number of steps of the Search Algorithm is at most $\sum_{H\in\mathfrak{S}} \mu^\top A_H \vec{1}$.

Proof. For each time $t$ that a flaw $f$ is resampled, the corresponding $G_t$ is an appearing wdag in $\mathfrak{W}(f)$. These wdags are all distinct, since on the $k^{\text{th}}$ resampling of $f$ the wdag $G_t$ has exactly $k$ nodes labeled $f$. So by Lemma 3.5, the expected number of steps is at most

$$\sum_f \sum_{H\in\mathfrak{W}(f)} \Pr(H \text{ appears}) \leq \sum_f \sum_{H\in\mathfrak{W}(f)} \mu^\top A_H \vec{1} = \sum_{H\in\mathfrak{S}} \mu^\top A_H \vec{1}. \qquad \square$$

## 3.2 Estimating sums over wdags

The statement of Lemma 3.5 in terms of matrix products is very general and powerful, but difficult for calculations. To use it effectively, as for example in Corollary 3.6, we need to bound the sums of the form

$$\sum_H \mu^\top A_H \vec{1}$$

where $H$ ranges over subsets of $\mathfrak{W}$.

There are two, quite distinct, issues here. First, for each individual wdag $H$, we need to estimate $\mu^\top A_H \vec{1}$; second, we need to bound the sum of these quantities over $H$. The second issue is well-studied in terms of the probabilistic LLL, but the first issue is not as familiar. To handle it, following [3], we analyze the matrix product via spectral bounds of the matrices $A_f$. Let us define a quantity called the weight $w(f)$ of each flaw $f$ by[1]

$$w(f) := \max_{\tau\in\Omega} \frac{\mu^\top A_f e_\tau}{\mu(\tau)}$$

[1]The article [3] uses a more general definition of charge, where the "benchmark" probability distribution can be different from the initial probability distribution $\mu$. This can be useful in showing convergence of the Search Algorithm for non-commutative resampling oracles. However, this more general definition does not seem useful for distributional properties and parallel algorithms in the context of commutative resampling oracles. Hence, we adopt the simpler definitions here.

i. e., the maximum possible inflation of a state probability (relative to its probability under $\mu$) incurred by (i) sampling a state $\sigma$ according to $\mu$; (ii) checking that flaw $f$ holds on $\sigma$; and then (iii) resampling flaw $f$ at $\sigma$.

Extending the definition, we define the weight of a wdag $H$ by

$$w(H) = \prod_{v \in H} w(L(v))$$

Lemma 3.7. For any event $E \subseteq \Omega$ we have $\mu^\top A_H e_E \leq \mu(E) \cdot w(H)$.

Proof. We can write $A_H = A_{f_1} \cdots A_{f_t}$ where $f_1, \ldots, f_t$ are the labels of the nodes of $H$. By definition, we have $\mu^\top A_f \preceq w(f)\mu^\top$ for any $f$. So:

$$\mu^\top A_H e_E = \mu^\top A_{f_1} \cdots A_{f_t} e_E \leq \mu^\top w(f_1) A_{f_2} \cdots A_{f_t} e_E \leq \cdots \leq w(f_1) \cdots w(f_t) \mu^\top e_E = w(H)\mu(E).$$

□

We say that a resampling oracle R is regenerating if $w(f) = \mu(f)$ for all $f$. This perfectly removes the conditional of the resampled flaw. The original Moser-Tardos algorithm, and extensions to other probability spaces, can be viewed in terms of regenerating oracles [20]. An equivalent formulation is that $\mu$ is a left-eigenvector of each matrix $A_f$, with associated eigenvalue $\mu(f)$, i. e.,

$$\forall f \qquad \mu^\top A_f = \mu(f) \cdot \mu^\top \tag{1}$$

From Lemma 3.7 (applied with $E = \Omega$) and Lemma 3.5 we get the following immediate corollary.

Corollary 3.8. Every wdag $H$ appears with probability at most $w(H)$.

In particular, if R is regenerating, then it appears with probability at most $\prod_{v \in H} \mu(L(v))$ (i. e., the usual Witness Tree Lemma).

We emphasize that we are not aware of any direct proof of Corollary 3.8; it seems necessary to first show the matrix bound of Lemma 3.5, and then project down to scalars.

In light of Lemma 3.7, we define for any flaw set $I$ the key quantities

$$\Psi(I) = \sum_{H \in \mathfrak{W}(I)} w(H), \qquad \overline{\Psi}(I) = \sum_{J \subseteq I} \Psi(J).$$

We write $\Psi(f) = \Psi(\{f\})$ for brevity. Note that $\Psi(I) = 0$ if $I$ is not a stable set. A useful and standard formula (see, e. g., [20, Claim 59]) is that for any set $I$ we have

$$\Psi(I) \leq \prod_{f \in I} \Psi(f), \qquad \overline{\Psi}(I) \leq \prod_{f \in I} (1 + \Psi(f)).$$

We also write $\Psi_{\mathcal{F}}, \overline{\Psi}_{\mathcal{F}}$ to indicate the role of the flaw set $\mathcal{F}$, if it is relevant.

We summarize here a few well-known bounds on these quantities, based on versions of LLL criteria.

Proposition 3.9. 1. (Symmetric criterion) Suppose that $w(f) \leq p$ and $|\overline{\Gamma}(f)| \leq d$ for parameters $p, d$ with $epd \leq 1$. Then $\Psi(f) \leq ew(f) \leq ep$ for all $f$.

2. (Neighborhood bound) Suppose that every $f$ has $\sum_{g \in \overline{\Gamma}(f)} w(g) \leq \frac{1}{4}$. Then $\Psi(f) \leq 4w(f)$ for all $f$.

3. (Asymmetric criterion) Suppose there is some function $x : \mathcal{F} \to [0, 1)$ with the property that

$$\forall f \qquad w(f) \leq x(f) \prod_{g \in \Gamma(f)} (1 - x(g)).$$

Then $\Psi(f) \leq \frac{x(f)}{1-x(f)}$ for all $f$.

4. (Restricted cluster-expansion) Suppose there is some function $\eta : \mathcal{F} \to [0, \infty)$ with the property that

$$\forall f \qquad \eta(f) \geq w(f) \cdot \sum_{\substack{I \subseteq \overline{\Gamma}(f) \\ I \text{ stable}}} \prod_{g \in I} \eta(g)$$

Then $\Psi(f) \leq \eta(f)$ for all $f$.

5. (Cluster-expansion) Suppose $\bowtie$ is a symmetric relation on $\mathcal{F}$ extending $\simeq$, i. e., $f \simeq g \Rightarrow f \bowtie g$, and there is some function $\eta : \mathcal{F} \to [0, \infty)$ with the property that

$$\forall f \qquad \eta(f) \geq w(f) \cdot \sum_{\substack{I \subseteq \mathcal{F} \\ g \bowtie f \text{ for all } g \in I \\ g_1 \not\bowtie g_2 \text{ for all distinct } g_1, g_2 \in I}} \prod_{g \in I} \eta(g)$$

Then $\Psi(f) \leq \eta(f)$ for all $f$.

Proof. For completeness, we briefly sketch the proofs.

For the cluster-expansion criterion, first observe that given a wdag $G$, we can topologically sort the nodes of $G$ and then add an additional edge, in the appropriate direction, for each pair of nodes $v, w$ with $L(v) \bowtie L(w)$. In this way, $\mathrm{sink}(G)$ becomes a set which is independent with respect to the

denser dependence relation $\bowtie$. Now use induction on the number of nodes to show that the total weight of all wdags whose sink nodes (under $\bowtie$) are labeled by $I$ is at most $\prod_{f \in I} \eta(f)$. Here, we use the fact that if wdag $H$ has sink nodes $v_1, \ldots, v_t$ labeled by a stable set $I$, then the sink nodes of $H - v_1 - \cdots - v_t$ are labeled by a subset of $\bigcup_{f \in I} \{g : g \bowtie f\}$.

For the restricted cluster-expansion criterion, apply the cluster-expansion criterion with $\bowtie$ being $\simeq$.

For the asymmetric criterion, apply the cluster-expansion criterion with $\eta(f) = \frac{x(f)}{1-x(f)}$

For the neighborhood bound criterion, apply the asymmetric criterion with $x(f) = 2w(f)$.

For the symmetric criterion, apply the restricted cluster-expansion criterion with $\eta(f) = ew(f)$. □

To emphasize the connection between various LLL-type bounds, our analysis of wdags, and the behavior of the Search Algorithm, we record the following results:

Proposition 3.10. Define the parameter

$$W = \sum_{f \in \mathcal{F}} \Psi(f) = \sum_{H \in \mathfrak{S}} w(H)$$

The expected number of total resamplings is at most $W$. Moreover, under the conditions of Proposition 3.9, we have the following bounds on $W$:

1. If the symmetric criterion holds, then $W \leq e \sum_f w(f) \leq O(p|\mathcal{F}|)$.
2. If the neighborhood-bound criterion holds, then $W \leq 4 \sum_f w(f) \leq O(|\mathcal{F}|)$.
3. If the asymmetric criterion holds, then $W \leq \sum_f \frac{x(f)}{1-x(f)}$.
4. If the cluster-expansion criterion holds, then $W \leq \sum_f \eta(f)$.

# 4 Simple distributional properties

One of the most important consequence of commutativity is that it allows us to bound the distribution of objects generated by the Search Algorithm. As a warm-up, we will use a construction similar to Section 3 for a "basic"

distributional bound. Later, in Section 7, we show tighter bounds through a more careful constructions of wdags.

Consider an event $E \subseteq \Omega$, and let $P(E)$ be the probability that $E$ holds in the output of the Search Algorithm. Define $\check{\Gamma}(E)$ to be the set of flaws $f$ which can cause $E$ to occur, i. e., there is a transition $\sigma' \notin E \xrightarrow{f} \sigma \in E$. Our main goal is to bound $P(E)$; typically, we will seek an upper-bound of the form $P(E) \leq (1+\epsilon)\Pr_\Omega(E)$, for some small value $\epsilon > 0$.

If $E$ occurs at some time $t$ in the Search Algorithm, we generate a wdag $G_t$ by initializing $G_t = \emptyset$ and then for each time $s = t-1, \ldots, 0$ with resampled flaw $f_s$, modifying $G_t$ according to the following rule:

- If $f_s \sim G_t$, or $G_t$ has a source node labeled $f_s$, or $f_s \in \check{\Gamma}(E)$, then prepend $f_s$ to $G_t$.
- Otherwise, do not modify $G_t$.

Note that $\mathrm{sink}(G_t) \subseteq \check{\Gamma}(E)$. We say that wdag $H$ appears for $E$ if $H$ is isomorphic to $G_t$ for some $t$.

Lemma 4.1. The probability that wdag $H$ appears for $E$ is at most $\mu^\top A_H e_E$.

Proof. As in Lemma 3.5, it suffices to show that if the Search Algorithm runs for at most $t_{\max}$ steps starting with state $\sigma$, where $t_{\max}$ is an arbitrary integer, then $H$ appears with probability at most $e_\sigma^\top A_H e_E$. We prove this claim by induction on $t_{\max}$.

If $H = \emptyset$ and $\sigma \in E$, then $e_\sigma^\top A_H e_E = 1$ and the bound holds vacuously. This is the only way that $H$ can appear if $t_{\max} = 0$. So, for the induction step, suppose that $t_{\max} > 0$ and either $H \neq \emptyset$ or $\sigma \notin E$. Then $G_0 \neq \emptyset$, and the only way for $H$ to appear is to have $G_t = H$ for $t \in \{1, \ldots, t_{\max}\}$. Now suppose S selects a flaw $f$ to resample in $\sigma$. We can view the evolution of the Search Algorithm A as a two-part process: first resample $f$, reaching a state $\sigma'$; then execute a new search algorithm A$'$ starting at state $\sigma'$.

As in Lemma 3.5, in order for $H$ to appear, one of the following two mutually exclusive conditions must hold: (i) $H$ has a unique source node $v$ labeled $f$ and $G'_{t-1} = H - v$; or (ii) $f \not\sim H$ and $f \notin \check{\Gamma}(E)$ and $G'_{t-1} = H$.

In case (i), $H - v$ would appear for $E$ in search algorithm A$'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most $e_{\sigma'}^\top A_{H-v} e_E$ for a fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of $\sum_{\sigma'} A_f[\sigma, \sigma'] e_{\sigma'}^\top A_{H-v} e_E = e_\sigma^\top A_f A_{H-v} e_E = e_\sigma^\top A_H e_E$.

In case (ii), $H$ would appear for $E$ in search algorithm A$'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most $e_{\sigma'}^\top A_H e_E$

for a fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of $\sum_{\sigma'} A_f[\sigma,\sigma']e_{\sigma'}^\top A_H e_E = e_\sigma^\top A_f A_H e_E$. Since $A_f$ commutes with $A_H$, this equals $e_\sigma^\top A_H A_f e_E$. Since $f \notin \check{\Gamma}(E)$, resampling $f$ can never cause $E$ to occur, and so $e_\tau^\top A_f e_E = 0$ for any state $\tau \notin E$; equivalently, we have $A_f e_E \preceq e_E$. So, overall, the probability is at most $e_\sigma^\top A_H e_E$ as claimed. □

This gives us the following crisp bound:

Theorem 4.2. $P(E) \le \mu(E)\overline{\Psi}(\check{\Gamma}(E))$.

Proof. If $E$ is ever true, then some wdag $H$ appears for $E$, where necessarily $\mathrm{sink}(H) \subseteq \check{\Gamma}(E)$. A union bound over such wdags gives $P(E) \le \sum_{I\subseteq\check{\Gamma}(E)} \sum_{H\in\mathfrak{W}(I)} \mu^\top A_H e_E$. By Lemma 3.7, this is at most $\sum_{I\subseteq\check{\Gamma}(E)} \sum_{H\in\mathfrak{W}(I)} w(H)\mu(E) = \mu(E)\overline{\Psi}(\check{\Gamma}(E))$. □

We can combine this with common LLL criteria to obtain more readily usable bounds; the proofs are immediate from bounds on $\Psi$ shown in Proposition 3.9.

Proposition 4.3. Under four criteria of Proposition 3.9, we have the following estimates for $P(E)$:

1. If the symmetric criterion holds, then $P(E) \le \mu(E) \cdot \mathrm{e}^{\mathrm{e}|\check{\Gamma}(E)|p}$.
2. If the neighborhood-bound criterion holds, then $P(E) \le \mu(E) \cdot \mathrm{e}^{4\sum_{f\in\check{\Gamma}(E)} w(f)}$.
3. If function $x$ satisfies the asymmetric criterion, then $P(E) \le \mu(E) \cdot \prod_{f\in\check{\Gamma}(E)} \frac{1}{1-x(f)}$.
4. If function $\eta$ satisfies the cluster-expansion criterion, then $P(E) \le \mu(E) \cdot \sum_{\substack{I\subseteq\check{\Gamma}(E)\\ I \text{ stable}}} \prod_{g\in I} \eta(g)$.

We remark that Iliopoulos [21] had shown a bound similar to Theorem 4.2, but with three additional technical restrictions: (i) it requires strong commutativity; (ii) it requires the construction of a commutative resampling oracle for the event $E$ itself; and (iii) it gives a strictly worse bound for non-regenerating oracles.

# 5 Parallel algorithms

Moser & Tardos [26] described a simple parallel version of their resampling algorithm:

Algorithm 3 Parallel Moser-Tardos algorithm

1: Draw state $X$ from distribution $\mu$
2: while some bad event is true on $X$ do
3: Select some arbitrary maximal independent set (MIS) $I$ of bad events true on $X$
4: Resample, in parallel, all variables $X_i$ involved in events in $I$

This algorithm depends heavily on the properties of the variable LLL, as it requires that bad events which are independent in the LLL sense are also computationally independent. Parallel algorithms have been developed for a number of other probability spaces [18, 13], with intricate and ad-hoc analyses. Based on commutativity, Kolmogorov [24] proposed a generic framework which we summarize as follows.

Algorithm 4 Generic parallel resampling framework

1: Draw state $\sigma$ from distribution $\mu$
2: while some flaw holds on $\sigma$ do
3: Set $V$ to be the set of flaws currently holding on $\sigma$
4: while $V \neq \emptyset$ do
5: Select, arbitrarily, a flaw $f \in V$.
6: Update $\sigma \leftarrow \mathrm{R}_\sigma(\sigma)$.
7: Remove from $V$ all flaws $g$ such that $\sigma \notin g$ or $f \simeq g$

We emphasize that this is a sequential algorithm, which can be viewed as a version of the Search Algorithm with an unusual flaw-selection rule. Each iteration of the main loop (lines 3 – 7) is called a round. Kolmogorov showed that if R is strongly commutative, then Algorithm 4 terminates after polylogarithmic rounds with high probability. Harris [15] further showed that if R satisfies a condition called obliviousness (see Section 6), then each round can be simulated in polylogarithmic time. These two results combine to give an overall RNC search algorithm. Most known parallel local search algorithms, including Algorithm 3, fall into this framework.

We will extend these results to the matrix-commutative setting. For $k = 1, 2, \ldots$, define $V_k$ to be the set of flaws $V$ in round $k$, and define $I_k$ to be the set of flaws which are actually resampled at round $k$ (i. e., a flaw $f$ selected at some iteration of line 5). Note that each $I_k$ is a stable set and $I_k \subseteq V_k$.

Let $b_k = \sum_{i<k} |I_i|$ be the total number of resamplings made before round $k$; thus $b_1 = 0$, and when we "serialize" Algorithm 4 and view it as an in-

stance of the Search Algorithm, the resamplings in round $k$ of Algorithm 4 correspond to the resamplings at times $b_k, \ldots, b_{k+1} - 1$ of the Search Algorithm.

Proposition 5.1. For each $f \in V_k, k \geq 2$ there exists $g \in I_{k-1}$ with $f \simeq g$.

Proof. First, suppose that $f \notin V_{k-1}$. In this case, by Observation 2.4, the only way $f$ could become true at round $k$ would be that some $g \sim f$ was resampled at round $k-1$, i. e., $g \in I_{k-1}$. Otherwise, suppose that $f \in V_{k-1}$. Then either it was removed from $V_{k-1}$ due to resampling of some $g \simeq f$, or $f$ became false during round $k-1$. In the latter case, since it later become true at the beginning of round $k$, some other $g' \sim f$ was resampled in round $k-1$ after $g$. □

For a node $v$ in a wdag $G$, we define the depth of $v$ to be the length of the longest path to any sink node; we define the depth of $G$ to be the maximum depth of its vertices.

Proposition 5.2. For each $t \in \{b_k, \ldots, b_{k+1} - 1\}$ the corresponding wdag $G_t$ as constructed in Section 3.1 has depth precisely $k$.

Proof. Fix $t$, and consider the partial construction of $G_t$ by adding nodes backwards in time for each flaw $f_s$ resampled at times $s = t, t-1, \ldots, b_j$; let $H_j$ be the resulting wdag. So $H_1 = G_t$. We show by backwards induction on $j$ that each $H_j$ has depth precisely $k - j + 1$, and the nodes $v \in H_j$ with depth $k - j + 1$ correspond to resamplings in round $j$.

The base case $j = k$ is clear, since then $H_j$ consists of a singleton node corresponding to the resampling at time $t$ in round $k$.

For the induction step, observe that $H_j$ is formed from $H_{j+1}$ by adding nodes corresponding to resamplings in $I_j$. By induction hypothesis, $H_{j+1}$ has depth $k-j$. Since $I_j$ is a stable set, we have $\text{depth}(H_j) \leq 1 + \text{depth}(H_{j+1}) = k - j + 1$ and furthermore the nodes at maximal depth correspond to resamplings in $I_j$. So we just need to show that there is at least one such node.

Consider any node $v$ of $H_{j+1}$ with depth $k - j$; by induction hypothesis this corresponds to a resampling in round $j+1$. By Proposition 5.1, we have $L(v) \simeq f_s$ for some time $s$ in round $j$. The procedure for generating $G_t$ will thus add a node labeled $f_s$ with an edge to $v$; this node has depth $k - j + 1$ in $H_j$.

This completes the induction. The stated bound then holds since $G_t = H_1$. □

Proposition 5.3. For any flaw $f$ and $k \geq 1$, we have $\Pr(f \in V_k) \leq \sum_{\substack{H \in \mathfrak{W}(f) \\ \text{depth}(H)=k}} \mu^\top A_H \vec{1}$.

Proof. As discussed, Algorithm 4 can be viewed as an instantiation of the Search Algorithm with a certain flaw selection rule S. For fixed $f$ and $k$, consider a new flaw selection rule $\mathrm{S}_{f,k}$, which agrees with S up to round $k$, and then selects $f$ to resample at round $k$ if it is true. The behavior of the Search Algorithm for S and $\mathrm{S}_{f,k}$ is identical up through the first $b_k$ resamplings. Subsequently, we have $f \in V_k$ if and only if the Search Algorithm with $\mathrm{S}_{f,k}$ selects $f$ at time $t = b_k$. In this case, by Proposition 5.2, the corresponding wdag $G_t \in \mathfrak{W}(f)$ would have depth $k$.

So we can bound the probability of $f \in V_k$ by a union bound over such wdags $H \in \mathfrak{W}(f)$ of depth $k$ and applying Lemma 3.5. □

As is usual for the parallel LLL, we analyze the running time by using an "inflated" weight function for some $\epsilon > 0$ defined as

$$w_\epsilon(H) = w(H)(1+\epsilon)^{|H|} = \prod_{v \in H} (1+\epsilon) w(L(v)) \qquad W_\epsilon = \sum_{H \in \mathfrak{S}} w_\epsilon(H)$$

Bounding $W_\epsilon$ is very similar to bounding $W = W_0$ with a small "slack" in the weights. With this definition, we get the following bounds:

Proposition 5.4. 1. $\sum_k \mathbb{E}[|V_k|] \leq W$.

2. For any integer $t \geq 1$ and any $\epsilon > 0$, the probability that Algorithm 4 runs for more than $2\ell$ rounds is at most $(1+\epsilon)^{-\ell} W_\epsilon / \ell$.

3. For $\epsilon, \delta \in (0, \frac{1}{2})$, Algorithm 4 terminates in $O(\frac{\log(1/\delta + \epsilon W_\epsilon)}{\epsilon})$ rounds with probability at least $1 - \delta$.

Proof. First, by Lemma 3.7 and Proposition 5.3, we have

$$\mathbb{E}[|V_k|] \leq \sum_f \sum_{\substack{H \in \mathfrak{W}(f), \\ \text{depth}(H)=k}} w(H) = \sum_{\substack{H \in \mathfrak{S}, \\ \text{depth}(H)=k}} w(H),$$

so $\sum_k \mathbb{E}[|V_k|] \leq \sum_{H \in \mathfrak{S}} w(H) = \sum_f \Psi(f) = W$.

For the second claim, consider random variable $Y = \sum_{k \geq \ell} |V_k|$. If Algorithm 4 reaches iteration $2\ell$, then necessarily $V_k \neq \emptyset$ for $k = \ell, \dots, 2\ell$, and so $Y \geq \ell$. By Markov's inequality applied to $Y$, we thus get

$$\Pr(\text{Alg reaches round } 2\ell) \leq \Pr(Y \geq \ell) \leq \mathbb{E}[Y]/\ell \leq \sum_{H \in \mathfrak{S}, \text{depth}(H) \geq \ell} w(H)/\ell.$$

We can then calculate

$$\sum_{H\in\mathfrak{S},\mathrm{depth}(H)\geq\ell} w(H) = \sum_{H\in\mathfrak{S},\mathrm{depth}(H)\geq\ell} w_\epsilon(H)(1+\epsilon)^{-|H|} \leq (1+\epsilon)^{-\ell} \sum_{H\in\mathfrak{S}} w_\epsilon(H) = (1+\epsilon)^{-\ell} W_\epsilon.$$

The third claim follows from the second claim via Markov's inequality. □

Using standard estimates (see [11, 24, 3]) we get the following bounds:

Proposition 5.5.
1. If the resampling oracle is regenerating and the probability vector $p(1+\epsilon)$ satisfies the LLLL criterion, then $W_{\epsilon/2} \leq O(m/\epsilon)$ and Algorithm 4 terminates after $O(\frac{\log(m/\delta)}{\epsilon})$ rounds with probability $1-\delta$.

2. If $w(f) \leq p$ and $|\overline{\Gamma}(f)| \leq d$ for alls flaws $f$, where $epd(1+\epsilon) \leq 1$, then $W_{\epsilon/2} \leq O(m/\epsilon)$ and Algorithm 4 terminates after $O(\frac{\log(m/\delta)}{\epsilon})$ rounds with probability at least $1-\delta$.

3. If the resampling oracle is regenerating and oblivious and satisfies the computational requirements of [15] for input length $n$, then with probability $1-1/\operatorname{poly}(n)$ the algorithm of [15] terminates in $O(\frac{\log^4(n+\epsilon W_\epsilon)}{\epsilon})$ time on an EREW PRAM.

# 6 Compositional properties for resampling oracles

In many settings, the flaws and resampling oracles are built out of simpler, "atomic" events. In [15], Harris described a generic construction, and generic parallel algorithm, when the atomic events satisfy an additional condition called obliviousness. Let us now review this construction, and how it works with commutativity.

Consider a set $\mathcal{A}$ of events, and a symmetric dependence relation $\sim$. It is allowed, but not required, to have $f \sim f$ for $f \in \mathcal{A}$. We refer to the elements of $\mathcal{A}$ as atoms. These should be thought of as "pre-flaws", that is, they have the structural properties of a resampling oracle, but do not necessarily satisfy any convergence condition such as the LLLL.

The obliviousness definition in [15] can be formulated as follows.

Definition 6.1 (Explicitly oblivious resampling oracle [15]). The resampling oracle R is explicitly oblivious if each $\mathrm{R}_f$ can be implemented by drawing a random seed $r$ from a set $R_f$ and setting $\sigma' = \mathrm{R}_f(\sigma) = F_f(\sigma, r)$ for a

deterministic function $F_f$. Furthermore, for each pair $f, g \in \mathcal{A}$ with $f \not\sim g$, there is a set $R_{f;g} \subseteq R_f$ such that

$$(F_f(\sigma, r) \in g) \Leftrightarrow (\sigma \in f \cap g \wedge r \in R_{f;g})$$

We list a number of probability spaces with this property; see [15] for further details and proofs.

1. Variables: This is easy. The probability space $\Omega$ has $n$ independent variables $X_1, \ldots, X_n$. For each pair $(i, y)$, we have an atom $f_{iy} = \{X_i = y\}$. We have $f_{iy} \sim f_{i'y'}$ iff $i = i', y \neq y'$. The space $R_{f_{iy}}$ is defined by drawing a value $y'$ from the distribution of $X_i$, and setting $F_f(X, y') = (X_1, \ldots, X_{i-1}, y', X_{i+1}, \ldots, X_n)$.

2. Permutations: $\Omega$ is the uniform distribution on permutations $\pi$ on $\{1, \ldots, n\}$. For each pair $(x, y)$, we have an atom $f_{xy} = \{\pi x = y\}$. We have $f_{xy} \sim f_{x'y'}$ iff $x = x', y \neq y'$ or $x \neq x', y = y'$. The space $R_{f_{xy}}$ is defined by choosing a value $z$ uniformly from $\{1, \ldots, n\}$, and setting $F(\pi, z) = (y\ z)\pi$. (Here $(y\ z)$ is the permutation which swaps $y$ with $z$.)

3. Clique perfect matchings: $\Omega$ is the uniform distribution on perfect matchings $M$ of the $n$-clique, where $n$ is even. For each pair $(x, y)$, we have an atom $f_{xy} = \{\{x, y\} \in M\}$. Note that $f_{xy} = f_{yx}$. We have $f_{xy} \sim f_{x'y'}$ iff $|\{x, y\} \cap \{x', y'\}| = 1$. For $x < y$, the space $R_{f_{xy}}$ is defined by choosing a value $z$ uniformly from $\{1, \ldots, n\} \setminus \{x\}$, and setting $F(M, z) = (y\ z)M$ (with the natural left-group action of permutations on matchings).

4. Hamiltonian cycles: $\Omega$ is the uniform distribution on $n$-cycle permutations. For each sequence of distinct elements $\vec{x} = x_1, \ldots, x_k$ there is an atom $f_x = \{\pi x_i = x_{i+1} : i = 1, \ldots, k-1\}$. We have $f_x \sim f_{x'}$ iff $\{x_1, \ldots, x_k\} \cap \{x'_1, \ldots, x'_{k'}\} \neq \emptyset$. The resampling oracle for this space is too complicated to explain here.

While this definition of obliviousness is critical for the parallel algorithm, we use a simpler and more general matrix-based notion.

Definition 6.2 (Matrix-oblivious resampling oracle). The resampling oracle R is matrix-oblivious if for each stable set $C$ and atom $f \not\sim C$ there holds $A_f e_{\langle C \rangle} \propto e_{f \cap \langle C \rangle}$.

(Recall that $\langle C \rangle$ is the intersection of the atoms in $C$.)

Proposition 6.3. If R is explicitly oblivious but not necessarily commutative, then R is matrix-oblivious.

Proof. Let $C = \{g_1, \ldots, g_k\}$ and define $g = \langle C \rangle$. Observe that, when implementing $\sigma' = \mathrm{R}_f(\sigma)$ by drawing a seed $r$ from $R_f$, we have $\sigma' \in g$ if and only if $r \in R_{f;g_1} \cap \cdots \cap R_{f;g_k}$ and $\sigma \in f \cap g_1 \cap \cdots \cap g_k = f \cap g$. In particular, $e_\sigma^\top A_f e_g$ is constant for $\sigma \in f \cap g$ and hence $A_f e_g \propto e_{f \cap g}$. □

For the remainder of this paper, we say oblivious to mean matrix-oblivious.

Let us suppose now that R is oblivious. From $\mathcal{A}$, one can construct an enlarged set of events

$$\overline{\mathcal{A}} = \{\langle C \rangle \mid C \text{ a stable set of } \mathcal{A}\}.$$

We define the relation $\sim$ on $\overline{\mathcal{A}}$ by setting $\langle C \rangle \sim \langle C' \rangle$ iff $C \sim C'$. For each event $f = \langle C \rangle \in \overline{\mathcal{A}}$, we define a corresponding resampling oracle $\overline{\mathrm{R}}_f$ as follows. Choose some arbitrary enumeration $C = \{f_1, \ldots, f_t\}$. Given a state $\sigma_1$, apply $\mathrm{R}_{f_1}$ repeatedly until the resulting state $\sigma_2$ is in $f_2 \cap \cdots \cap f_t$ (i. e., via rejection sampling). Then, again apply $\mathrm{R}_{f_2}$ repeatedly until the state is in $f_3 \cap \cdots \cap f_t$, and so on.

The intent is to choose the flaw set $\mathcal{F}$ to be some arbitrary subset of $\overline{\mathcal{A}}$; as before, $\overline{\mathcal{A}}$ does not necessarily satisfy any LLLL convergence criterion.

It would seem reasonable that if R is commutative, then $\overline{\mathrm{R}}$ should be as well. We will indeed show this for matrix commutativity. By contrast, it does not seem to hold for strong commutativity. This is a good illustration of how the new commutativity definition is easier to work with, beyond its advantage of greater generality.

Proposition 6.4. Suppose R is oblivious and regenerating, but not necessarily commutative. Then for a stable set $C$ and atom $f \not\sim C$ there holds $A_f e_{\langle C \rangle} = \frac{\mu(f)\mu(\langle C \rangle)}{\mu(f \cap \langle C \rangle)} e_{f \cap \langle C \rangle}$.

Proof. By hypothesis, we have $A_f e_{\langle C \rangle} = p \cdot e_{f \cap \langle C \rangle}$ for scalar $p$. Since R is regenerating, we have on the one hand $\mu^\top A_f e_{\langle C \rangle} = \mu(f)\mu^\top e_{\langle C \rangle} = \mu(f)\mu(\langle C \rangle)$, and on the other hand $\mu^\top e_{f \cap \langle C \rangle} = \mu(f \cap \langle C \rangle)$. □

Proposition 6.5. Suppose R is oblivious but not necessarily commutative. For $f = \langle C \rangle$ in $\overline{\mathcal{A}}$, suppose that we have fixed an enumeration $C = \{f_1, \ldots, f_t\}$ to define $\overline{\mathrm{R}}_f$. Then $A_f \propto A_{f_1} \cdots A_{f_t}$. If R is regenerating, then in particular $A_f = \frac{\mu(f)}{\mu(f_1) \cdots \mu(f_t)} A_{f_1} \cdots A_{f_t}$.

Proof. We show it by induction on $t$. The case $t = 1$ is immediate. For $t > 1$, let $f' = \langle\{f_2, \ldots, f_t\}\rangle$. We can view $\overline{\mathrm{R}}_f(\sigma)$ recursively: first resample $f_1$, conditional on the resulting state $\sigma'$ being in $f'$; then apply $\mathrm{R}_{f'}$. So $e_\sigma^\top A_f = \frac{e_\sigma^\top A_{f_1} A_{f'}}{e_\sigma^\top A_{f_1} e_{f'}}$. By Proposition 6.3 we have $e_\sigma^\top A_{f_1} e_{f'} \propto e_\sigma^\top e_{f_1 \cap f'} = 1$ since $\sigma \in f_1 \cap f' = f$. Overall, we have $A_f \propto A_{f_1} A_{f'}$, and the first result then follows immediately from induction.

The proof for the case where R is regenerating is completely analogous, where we use Proposition 6.4 to determine $e_\sigma^\top A_{f_1} e_{f'} = \mu(f_1)\mu(f')/\mu(f)$ for $\sigma \in f$. □

Theorem 6.6. Suppose R is oblivious, but not necessarily commutative.

- $\overline{\mathrm{R}}$ with dependence relation $\sim$ is an oblivious resampling oracle for $\overline{\mathcal{A}}$.
- If R is regenerating on $\mathcal{A}$, then $\overline{\mathrm{R}}$ is regenerating on $\overline{\mathcal{A}}$.

Proof. Consider $f = \langle\{f_1, \ldots, f_t\}\rangle \in \overline{\mathcal{A}}$ and a stable set $C = \{\langle C_1\rangle, \ldots, \langle C_k\rangle\}$ of $\overline{\mathcal{A}}$ with $f \not\sim C$. Let $g = \langle C\rangle$; note that also $g = \langle C_1 \cup \cdots \cup C_k\rangle$, where $C_1 \cup \cdots \cup C_k$ is a stable set of $\mathcal{A}$. By Proposition 6.5, we have $A_f e_g \propto A_{f_1} \cdots A_{f_t} e_g$. By matrix-obliviousness of $f_t$, this is proportional to $A_{f_1} \ldots A_{f_{t-1}} e_{f_t \cap g}$. Repeatedly again applying the definition of matrix-oblivious to atoms $f_{t-1}, \ldots, f_1$ gives us us $A_f e_g \propto e_{f_1 \cap \cdots \cap f_t \cap g} = e_{f \cap g}$ as required for the first claim. Similarly, if R is regenerating, then Proposition 6.5 gives $\mu^\top A_g = \frac{\mu^\top \mu(g)}{\mu(f_1)\cdots\mu(f_t)} A_{f_1} \cdots A_{f_t} = \mu(g)\mu^\top$ since $\mu^\top A_{f_i} = \mu(f_i)$ for each $i$. □

Theorem 6.7. Suppose R is commutative and oblivious. Then the following properties hold for $\overline{\mathrm{R}}$:

1. For a stable set $I = \{\langle C_1\rangle, \ldots, \langle C_k\rangle\}$ in $\overline{\mathcal{A}}$, the multiset union $J = C_1 \uplus C_2 \uplus \cdots \uplus C_k$ (i. e., the number of copies of $f$ is the sum of those in $C_1, \ldots, C_k$) is a stable multiset of $\mathcal{A}$, with $A_I \propto A_J$.
2. The matrix $A_f$ for $f = \langle C\rangle$ in $\overline{\mathcal{A}}$ does not depend on the chosen enumeration $C = \{f_1, \ldots, f_t\}$.
3. $\overline{\mathrm{R}}$ is commutative on $\overline{\mathcal{A}}$.

Proof. 1. Since $I$ is stable in $\overline{\mathcal{A}}$, we have $C_i \not\sim C_j$ for all pairs $i, j$. So $J$ is a stable multiset. Furthermore, by Proposition 6.5, we have we have $A_I = \prod_{i=1}^k A_{\langle C_i\rangle} \propto \prod_{i=1}^k \prod_{f \in C_i} A_f = \prod_{f \in J} A_f = A_J$.

2. By Proposition 6.5, we have $A_f = cA_{f_1} \cdots A_{f_t}$ for a scalar $c$. Since the matrices $A_{f_i}$ all commute, the RHS does not depend on the enumeration of $C$. Furthermore, $c$ can be determined from $A_{f_1} \cdots A_{f_t}$ by choosing an arbitrary state $\sigma \in f$ and setting $c = \frac{1}{e_\sigma^\top A_{f_1} \cdots A_{f_t} \vec{1}}$.

3. Let $g = \langle C \rangle, g' = \langle C' \rangle$ with $g \not\sim g'$. So $f \not\sim f'$ for all $f \in C, f' \in C'$. By Proposition 6.5 we have

$$A_g A_{g'} = c_g c_{g'} \Big( \prod_{f \in C} A_f \prod_{f' \in C'} A_{f'} \Big), \qquad A_{g'} A_g = c_{g'} c_g \Big( \prod_{f' \in C'} A_{f'} \prod_{f \in C} A_f \Big)$$

for scalar constants $c_g, c_{g'}$. All these matrices $A_f, A_{f'}$ commute, so both quantities are equal. □

As an example of an LLL construction using atomic events, recall the latin transversal application. Here, we have an $n \times n$ colored array $C$, where each color appears at most $\Delta$ times. We seek a permutation $\pi$ where all colors $C(i, \pi)$ are distinct.

Proposition 6.8. Suppose that $\Delta = \frac{27}{256} n$. Then the expected number of steps of the Search Algorithm is $O(n)$. Furthermore, we have $\Psi(f) \leq \frac{256}{81n^2}$ for each flaw $f$.

Proof. We start with the atomic set $\mathcal{A}$ for the permutation setting, and then build the set of flaws $\mathcal{F}$ as a subset of $\overline{\mathcal{A}}$. For each pair of cells $(x_1, y_1), (x_2, y_2)$ of the same color, we have a flaw $f = \langle \{ f_{x_1 y_1}, f_{x_2 y_2} \} \rangle$, i. e., that $\pi x_1 = y_1 \wedge \pi x_2 = y_2$.

We apply the cluster-expansion criterion with $\eta(f) = \frac{256}{81n^2}$ for each flaw $f$ and define the $\bowtie$ relation by setting $f \bowtie g$ if $f, g$ overlap in a row or column. This indeed extends $\simeq$, which has $f \sim g$ if they overlap on a coordinate and disagree on the corresponding other coordinate.

Consider now a flaw $f$ corresponding to cells $(x_1, y_1), (x_2, y_2)$, and a corresponding set $I$ of $\bowtie$-neighbors. At most one flaw $g \in I$ can overlap with column $x_1$ (any two such elements $g_1, g_2$ would have $g_1 \bowtie g_2$); given $x_1' = x_1$, there are $n$ choices for $y_1'$, then given the pair $(x_1', y_1')$, there are at most $\Delta - 1$ other cells with the same color. Similar arguments apply to elements in $I$ overlapping the other rows and columns. Overall, the sum of $\prod_{g \in I} \eta(g)$ over all such sets $I$ is at most $(1 + n(\Delta - 1)\frac{256}{81n^2})^4$. So we need to show that

$$\frac{256}{81n^2} \geq \frac{1}{n^2} \cdot \big(1 + n(\Delta - 1)\frac{256}{81n^2}\big)^4$$

which is a routine calculation for $n \geq 2$.

The total number of flaws is at most $n^2(\Delta-1)/2 = O(n^3)$. So $W \leq |\mathcal{F}| \cdot \frac{256}{81n^2} \leq O(n)$. □

# 7 More detailed distributional bounds

As before, consider an event $E$ in $\Omega$, and let $P(E)$ be the probability that $E$ holds in the output of the Search Algorithm. We will develop tighter bounds on $P(E)$, via a more refined construction of wdags. Namely, suppose that $E$ holds at some time $t$. We build a corresponding wdag $J_t$ by initializing $J_t = \emptyset$ and then, for each time $s = t-1, \ldots, 0$ with resampled flaw $f_s$, updating $J_t$ as follows.

- If $A_{f_s} A_{J_t} e_E \not\preceq A_{J_t} e_E$, or if $J_t$ has a source node labeled $f_s$, then prepend $f_s$ to $J_t$
- Otherwise, do not modify $J_t$.

We say that wdag $H$ appears for $E$ if $J_t$ is isomorphic to $H$ for any time $t \geq 0$. (This overrides the definition in Section 4.)

Lemma 7.1. Any given wdag $H$ appears for $E$ with probability at most $\mu^\top A_H e_E$.

Proof. As in Lemma 3.5, it suffices to show that if the Search Algorithm runs for at most $t_{\max}$ steps starting with state $\sigma$, where $t_{\max}$ is an arbitrary integer, then $H$ appears for $E$ with probability at most $e_\sigma^\top A_H e_E$. We prove this claim by induction on $t_{\max}$.

If $H = \emptyset$ and $\sigma \in E$, then $e_\sigma^\top A_H e_E = 1$ and the bound holds vacuously. This is the only way that $H$ can appear if $t_{\max} = 0$. So, for the induction step, suppose that $t_{\max} > 0$ and either $H \neq \emptyset$ or $\sigma \notin E$. Then $J_0 \neq H$, and the only way for $H$ to appear for $E$ is to have $J_t = H$ for some $t \in \{1, \ldots, t_{\max}\}$. Now suppose S selects a flaw $f$ in $\sigma$. We view the evolution of the Search Algorithm A as a two-part process: we first resample $f$, reaching state $\sigma'$ with probability $A_f[\sigma, \sigma']$. We then execute a new search algorithm A$'$ starting at state $\sigma'$.

So in order for $H$ to appear, one of the following two mutually exclusive conditions must hold: (i) $H$ has a unique source node $v$ labeled $f$ and $J'_{t-1} = H - v$; or (ii) $H$ has no such node and $J'_{t-1} = J_t = H$ and $A_f A_H e_E \preceq A_H e_E$.

In case (i), $H - v$ would appear for $E$ in search algorithm A$'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most

$e_{\sigma'}^\top A_H e_E$ for fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of

$$\sum_{\sigma'} A_f[\sigma, \sigma'] e_{\sigma'}^\top A_{H-v} e_E = e_\sigma^\top A_f A_{H-v} e_E = e_\sigma^\top A_H e_E$$

In case (ii), $H$ would appear for $E$ in search algorithm $\mathrm{A}'$ within $t_{\max} - 1$ timesteps. By induction hypothesis, this has probability at most $e_{\sigma'}^\top A_H e_E$ for fixed $\sigma'$. Summing over $\sigma'$ gives a total probability of

$$\sum_{\sigma'} A_f[\sigma, \sigma'] e_{\sigma'}^\top A_H e_E = e_\sigma^\top A_f A_H e_E$$

Since $A_f A_H \preceq A_H$, this is at most $e_\sigma^\top A_H e_E$, again completing the induction. □

Let us say that a time $t$ is good for $E$ if $E$ is true at time $t$, and either (i) $t = 0$ (i. e., the initial sampling of the variables) or (ii) $t > 0$ and $E$ is false at time $t - 1$. For a subset $X \subseteq \Omega$, we say that $t$ is good for $X, E$ if $t$ is good for $E$ and $X$ is true at time $0, \ldots, t - 1$. Define $N(X, E)$ to be the expected number of times $t$ that are good for $X, E$. Correspondingly, define $\mathfrak{J}[X, E]$ to be the collection of wdags which can appear for $E$ at such times $t$. We have the following main characterization:

Proposition 7.2. There holds $N(X, E) \leq \sum_{H \in \mathfrak{J}[X,E]} \mu^\top A_H e_E$.

Proof. We claim that, if $E$ is true at times $t_1, t_3$ and false at time $t_2$, where $t_1 < t_2 < t_3$, then $J_{t_1} \neq J_{t_3}$. We show this by induction on $t_1$. First suppose $t_1 = 0$. Then $J_{t_1} = \emptyset$. Since $E$ is false in A at time $t_2$, it must become true due to resampling a flaw $g$ at some time $t' \geq t_2$. Clearly $g \in \check{\Gamma}(E)$, i. e., $A_g e_E \not\preceq e_E$. So either $J_{t_3}$ is non-empty at time $t'$, or the rule for forming $J_{t_3}$ at time $t'$ would add a node labeled $g$ to the empty wdag. In either case we have $J_{t_3} \neq \emptyset = J_{t_1}$.

Next, suppose $t_1 > 0$ and $J_{t_1} = J_{t_3}$. Let $f_0$ be the flaw resampled at time 0, and consider the search algorithm $\mathrm{A}'$ starting at time 1, with corresponding wdags $J'_{t_1-1}$ and $J'_{t_3-1}$ where $E$ was false at time $t_2 - 1$. By induction hypothesis we have $J'_{t_1-1} \neq J'_{t_3-1}$. The only way to get $J_{t_1} = J_{t_3}$ is if we prepend $f_0$ to exactly one of $J'_{t_1-1}$ or $J'_{t_3-1}$. Say it is the former (the two cases are completely symmetric). So $J_{t_1} = J_{t_3} = J'_{t_3-1}$ would have a source node labeled $f_0$. But our rule for forming wdags would then also prepend $f_0$ to $J'_{t_3-1}$ to get $J_{t_3}$, a contradiction.

Thus, for each time $t$ that is good for $X, E$, some distinct wdag $J_t$ appears for $E$. By Lemma 7.1, the expected number of such times $t$ is at most $\sum_{H \in \mathfrak{J}[X,E]} \mu^\top A_H e_E$. □

Corollary 7.3. There holds $P(E) \leq \mu(E) \sum_{H \in \mathfrak{J}[\Omega \setminus E, E]} w(H)$.

Proof. Consider the first time $t$ that $E$ becomes true, if any. Then $E$ is false at times $0, \ldots, t-1$ and so $t$ is good for $\Omega \setminus E, E$. Hence, the expected number of such times is at most $N(\Omega \setminus E, E)$. This is at most $\sum_{H \in \mathfrak{J}[\Omega \setminus E, E]} \mu^\top A_H e_E$, which is at most $\mu(E) \sum_{H \in \mathfrak{J}[\Omega \setminus E, E]} w(H)$ via Lemma 3.7. □

To get improved bounds of $P(E)$ via Corollary 7.3, we need to analyze the wdags in $\mathfrak{J}[X, E]$ for the given event $E$ and for $X = \Omega \setminus E$. As a starting point, we observe two simple properties of such wdags.

Observation 7.4. For a wdag $H \in \mathfrak{J}[X, E]$, we have $\text{sink}(H) \subseteq \check{\Gamma}(E)$. Furthermore, for each node $v \in H$, we have $L(v) \cap X \neq \emptyset$.

Proof. For the first observation, suppose that we are building the wdag $J_t$ for a given time $t$ which is good for $X, E$, and suppose we add a sink node $v$ with label $f \notin \check{\Gamma}(E)$ at time $s < t$. Let $H$ denote the partial wdag before adding this node. Thus $f \not\sim H$, so $A_f A_H e_E = A_H A_f e_E$. Since $f \notin \check{\Gamma}(E)$, we have $A_f e_E \preceq e_E$. So $A_f A_H e_E \preceq A_H e_E$ and we would not add node $v$ to $J_t$.

For the second observation, note that if $t$ is good for $X, E$, then by definition $X$ holds at all previous times. Since the nodes of $J_t$ are labeled by previously seen flaws, we have $L(v) \cap X \neq \emptyset$ for any $v \in J_t$. □

In light of Observation 7.4, Corollary 7.3 gives bounds on $P(E)$ which are at least as tight as the simpler bound of Theorem 4.2. For some probability spaces, additional restrictions on the structure of $\mathfrak{J}[X, E]$, and tighter bounds of $P(E)$, can be available. These can be quite subtle and hard to analyze. The following characterization captures some (if not all) of these restrictions.

Proposition 7.5. For any wdag $H \in \mathfrak{J}[X, E]$, there is an enumeration $\text{sink}(H) = \{f_1, \ldots, f_k\}$ such that

$$\forall i = 1, \ldots, k \qquad A_{f_i} A_{f_{i+1}} \ldots A_{f_k} e_E \not\preceq A_{f_{i+1}} \ldots A_{f_k} e_E \tag{2}$$

Proof. Suppose the sink nodes of $J_t$ are added at increasing times $s_1, \ldots, s_k$, and let $f_i$ denote the flaw resampled at each time $s_i$. We claim that this enumeration satisfies the bound of Eq. (2). For, let $H$ denote the partial wdag $J_t$ produced immediately before prepending some $f_i$. Since $f_i$ should be the label of a sink node, we have $f_i \not\sim H$ and $\text{sink}(H) = \{f_{i+1}, \ldots, f_k\}$. We can write $A_H = A_{H'} A_{f_{i+1}} \cdots A_{f_k}$ where $H'$ are all the non-sink nodes

of $H$. By our rule for forming $J_t$, we must $A_{f_i} A_H e_E \not\preceq A_H e_E$. Using the fact that $A_{H'}$ and $A_{f_i}$ commute, this implies that $A_{H'} A_{f_i} A_{f_{i+1}} \cdots A_{f_k} e_E \not\preceq A_{H'} A_{f_{i+1}} \cdots A_{f_k} e_E$, which further implies that $A_{f_i} A_{f_{i+1}} \cdots A_{f_k} e_E \not\preceq A_{f_{i+1}} \cdots A_{f_k} e_E$ as claimed. □

To illustrate how this characterization can give significantly stronger bounds than Theorem 4.2 or known nonconstructive LLL estimates, we show the following result:

Theorem 7.6. In the variable, permutation, or clique-perfect-matching settings, let $C$ be a stable set of atomic events, and define event $E = \langle C \rangle$. For each $H \in \mathfrak{J}[X, E]$, there is an injective function $\phi_H : \text{sink}(H) \to C$ with $f \sim \phi_H(f)$ for all $f \in \text{sink}(H)$.

The article [13] showed a much stronger version of Theorem 7.6 for the variable setting. The article [14] showed (what is essentially) Theorem 7.6 for the permutation setting using a complicated and ad-hoc analysis based on a variant of witness trees. The bound for the clique-perfect-matching setting is new. We obtain all these results in a more unified way; the proofs are deferred to Appendix A.

Corollary 7.7. In the setting of Theorem 7.6, we have

$$N(X, E) \leq \mu(E) \prod_{g \in C} \big(1 + \sum_{f : f \sim g} \Psi_{\mathcal{G}}(f)\big) \qquad \text{for } \mathcal{G} = \{f \in \mathcal{F} : f \cap X \neq \emptyset\}.$$

Proof. First, by Observation 7.4, the nodes of any wdag $H \in \mathfrak{J}[X, E]$ are labeled from $\mathcal{G}$. Now let $C = \{g_1, \ldots, g_k\}$. To enumerate such $H$, we build the set $I = \text{sink}(H)$ by choosing, for each $i = 1, \ldots, k$, a preimage set $I_{g_i} = \phi_H^{-1}(g_i)$ of cardinality at most one. Given the choice of $I$, the remaining sum over wdags with $\text{sink}(H) = I$ is at most $\Psi_{\mathcal{G}}(I)$. Overall, this gives the bound:

$$\sum_I \Psi_{\mathcal{G}}(I) \leq \sum_{I_{g_1}, \ldots, I_{g_k}} \Psi_{\mathcal{G}}(I_{g_1} \cup \cdots \cup I_{g_k}) \leq \sum_{I_{g_1}, \ldots, I_{g_k}} \Psi_{\mathcal{G}}(I_{g_1}) \cdots \Psi_{\mathcal{G}}(I_{g_k})$$

where the last inequality follows from log-subadditivity of $\Psi$. This can be written as $\prod_{g \in C} \sum_{I_g} \Psi_{\mathcal{G}}(I_g)$. The case of $I_g = \emptyset$ contributes 1, and the case of $I_g = \{f\}$ contributes $\Psi_{\mathcal{G}}(f)$. □

We believe that tighter bounds on $N(X, E)$ are possible, using a more careful analysis of the structure of wdags in $\mathfrak{J}[X, E]$. Such bounds would lead to small improvements in the numerical constants for the later Theorem 7.11. Since the analysis needed to obtain Theorem 7.6 is already difficult, we leave this as an open problem.

## 7.1 A distributional bound with partial dependence

One weakness of the distributional bounds in Section 4 is that the definition of $\check{\Gamma}(E)$ is binary: either flaw $f$ cannot possibly cause $E$, or every occurrence of $f$ must be tracked to determine if it caused $E$. The next results allow us to take account of flaws which can "partially" cause $E$.

For flaw $f$ and event $E$, define

$$\kappa(f, E) = \max_{\sigma \in f \setminus E} \frac{e_\sigma^\top A_f e_E}{e_\sigma^\top A_f e_{E \cup (\Omega \setminus f)}}$$

Note that $\kappa(f, E) \in [0, 1]$, and $\kappa(f, E) = 0$ for $f \notin \check{\Gamma}(E)$. Thus, $\kappa(f, E)$ is a weighted measure of the extent to which $f$ causes $E$.

Theorem 7.8. $P(E) \leq \mu(E) + \sum_{f \in \mathcal{F}} \kappa(f, E) \cdot N(\Omega \setminus E, f \setminus E)$.

Proof. Let us say that a pair $(f, t)$ is open if the following three conditions hold: (i) $E$ is false at times $0, \ldots, t$; (ii) $f$ is resampled at time $t$; and (iii) either this is the first resampling of $f$, or if the most recent resampling of $f$ had occured at time $t' < t$, then $f$ had been false at some intermediate time between $t'$ and $t$. We say that a triple $(f, t, s)$ with $s \geq t$ is closed if $(f, t)$ is an open pair, and furthermore the following four conditions all hold: (i) $f$ is resampled at time $s$; (ii) $E$ has been false at all times up to $s$; (iii) the resampling at time $s$ make $E$ true; (iv) $f$ is true at times $t, \ldots, s$. Note that it is possible to have $s = t$.

We claim that if $E$ becomes true after time 0, then there is some closed triple $(f, t, s)$. For, suppose that $E$ first becomes true due to resampling flaw $f$ at time $s$. Going backward, let $t \leq s$ be the earliest time such that the same flaw $f$ is resampled at time $t$ and $f$ remained true between times $t$ and $s$. Then $(f, t)$ is open and $(f, t, s)$ is closed.

We next claim that, for a given pair $(f, t)$, the probability that there exists any closed triple $(f, t, s)$, conditional on that $(f, t)$ is open, is at most $\kappa(f, E)$. For, let $E' = E \cup (\Omega \setminus f)$, and we condition on the event that $s \geq t$ is the first time where $f$ is resampled and $E'$ becomes true. Let $\sigma \in f \setminus E$ be the state at time $s$. The probability that $E$ holds after resampling $f$, conditional on the state $\sigma$ and that $E'$ becomes true at that resampling, is $\frac{e_\sigma^\top A_f e_E}{e_\sigma^\top A_f e_{E'}}$, which is at most $\kappa(f, E)$ by definition.

Accordingly, the overall probability of $E$ is at most $\mu(E) + \sum_f L_f \cdot \kappa(f, E)$, where $L_f$ is the expected number of open pairs $(f, t)$, and the first term accounts for the possibility that $E$ holds at time 0. It remains to bound $L_f$. Let us fix some flaw $f$. We claim that, for each open pair $(f, t)$, there is

some distinct time $s_t$ which is good for $\Omega \setminus E, f \setminus E$. For, going backward, find the earliest time $s_t$ such that $f$ was true from times $s_t, \ldots, t$. If $s_t > 0$, then at time $s_t - 1$ the state transits from $\Omega \setminus f$ to $f \setminus E$, which is also a transition from $\Omega \setminus (f \setminus E)$ to $f \setminus E$. Furthermore, for distinct open pairs $(f, t), (f, t')$ we have $s_t \neq s_{t'}$. So $L_f \leq N(\Omega \setminus E, f \setminus E)$. □

Corollary 7.9. $P(E) \geq \mu(E) - \sum_{f \in \mathcal{F}} \kappa(f, \Omega \setminus E) \cdot N(E, f \cap E)$.

Proof. Apply Theorem 7.8 to the event $\Omega \setminus E$. □

For certain atomically generated resampling spaces, generic bounds on $\kappa$ are available.

Proposition 7.10. Let $\mathcal{A}$ be a commutative, oblivious, regenerating resampling space, which satisfies the following additional "strong obliviousness" condition[2]:

$$A_f e_g = \mu(g) e_f \qquad \text{for all atoms } f, g \in \mathcal{A} \text{ with } f \simeq g.$$

(This property holds, for example, for the variable, permutation, and clique-perfect-matching settings.)

Then for any event $f \in \overline{\mathcal{A}}$ and atom $g \in \mathcal{A}$ there holds

$$\kappa(f, g) \leq \frac{\mu(g)}{1 - \mu(f)}.$$

Moreover, if $f \not\sim g$, we have

$$\kappa(f, \Omega \setminus g) \leq \frac{1 - \frac{\mu(f)\mu(g)}{\mu(f \cap g)}}{1 - \mu(f)}$$

Proof. Let $E = g$ and $E = \Omega \setminus g$ in the two cases respectively. In either case, we can estimate the denominator in $\kappa$ as $e_\sigma^\top A_f e_{E \cup (\Omega \setminus f)} \geq e_\sigma^\top A_f e_{\Omega \setminus f} = 1 - e_\sigma^\top A_f e_f$. We claim that, for any such event $f = \langle \{f_1, \ldots, f_k\} \rangle$ and state $\sigma \in f$, there holds

$$e_\sigma^\top A_f e_f \leq \mu(f) \tag{3}$$

We show Eq. (3) by induction on $k$. The case $k = 1$ holds by hypothesis. For $k > 1$, let $f' = \langle \{f_2, \ldots, f_k\} \rangle$, and let $\sigma \in f$. By Proposition 6.5 and

[2]Postpublication correction (Nov. 23, 2025): erroneous term $\mu(f)e_g$ was replaced by $\mu(g)e_f$ in the strong obliviousness condition.

some manipulations can write:

$$e_\sigma^\top A_f e_f = \frac{\mu(f)}{\mu(f')\mu(f_1)} e_\sigma^\top A_{f_1} A_{f'} e_f \le \frac{\mu(f)}{\mu(f')\mu(f_1)} \sum_{\sigma' \in f_1} e_\sigma^\top A_{f_1} e_{\sigma'} \cdot e_{\sigma'}^\top A_{f'} e_{f'}$$

$$\le \frac{\mu(f)}{\mu(f')\mu(f_1)} \sum_{\sigma' \in f_1} e_\sigma^\top A_{f_1} e_{\sigma'} \mu(f') = \frac{\mu(f)}{\mu(f_1)} e_\sigma^\top A_{f_1} e_{f_1}. \qquad \text{(induction hypothesis)}$$

By hypothesis, is equal to $\mu(f)$.

Thus, the denominator in $\kappa$ (in either case) is at least $1 - \mu(f)$. We turn to estimating the numerator. For $\kappa(f, g)$, we claim that $e_\sigma A_f e_g = \mu(g)$ for any state $\sigma \in f$. For, consider $f = \langle \{f_1, \ldots, f_k\} \rangle$; by Theorem 6.7, any reordering of $f_1, \ldots, f_k$ would give the same matrix $A_f$. So assume without loss of generality that $f_k \sim g$. When implementing $\mathrm{R}_f$, suppose we condition on the state $\sigma'$ just before resampling $f_k$. By hypothesis, the probability that $\sigma'$ gets mapped to $g$ is precisely $\mu(g)$.

For $\kappa(f, \Omega \setminus g)$ with $g \not\sim f$, Proposition 6.4 gives

$$e_\sigma^\top A_f e_E = 1 - e_\sigma^\top A_f e_g = 1 - e_\sigma^\top \frac{\mu(f)\mu(g)}{\mu(f \cap g)} e_{f \cap g} = 1 - \frac{\mu(f)\mu(g)}{\mu(f \cap g)}. \qquad \square$$

## 7.2 Applications

Using the more sophisticated distributional bounds we can show the following bounds for the permutation and perfect-matching probability spaces.

Theorem 7.11. 1. If each color appears at most $\Delta = \frac{27}{256} n$ times in the array, then the Search Algorithm generates a latin transversal where, for each cell $x, y$, there holds

$$\frac{17}{32n} \le P(\pi x = y) \le \frac{203}{128n}$$

2. Consider an edge-coloring $C$ of the clique $K_n$, for $n$ even, such that each color appears on at most $\Delta = \frac{27}{256} n$ edges. Then the Search Algorithm generates a perfect matching $M$ such that $C(e) \neq C(e')$ for all distinct edges $e, e'$ of $M$. Moreover each edge $e$ has

$$\frac{17}{32(n-1)} \le P(e \in M) \le \frac{203}{128(n-1)}.$$

Proof. We show only the result on permutations; the result for the clique is completely analogous.

Let $g$ be the atomic event $g$ that $\pi x = y$. Note that any time that is good for $\Omega \setminus g, f \setminus g$ is good for $\Omega, f$ as well, so $N(\Omega \setminus g, f \setminus g) \leq N(\Omega, f)$. Thus, for the upper bound, Theorem 7.8 gives:

$$P(\pi x = y) \leq \mu(g) + \sum_{f \in \mathcal{F}} N(\Omega, f)\kappa(f, g)$$

Now consider a flaw $f \sim g$ defined by atoms $f_1, f_2$. By Proposition 7.10, we have

$$\kappa(f, g) \leq \frac{1/n}{1 - \frac{1}{n(n-1)}}$$

By Corollary 7.7, we have $N(\Omega, f) \leq \frac{(n-2)!}{n!} \prod_{i=1}^{2} \big(1 + \sum_{f' \in \mathcal{F}: f' \sim f_i} \Psi(f')\big)$. Since $\Psi(f') \leq \gamma := \frac{256}{81n^2}$, and for each $i = 1, 2$ there are at most $2n(\Delta - 1)$ choices for $f'$, we overall get $N(\Omega, f) \leq \frac{(1+2n(\Delta-1)\gamma)^2}{n(n-1)}$. There are at most $2n(\Delta - 1)$ choices for $f$, so we have

$$P(g) \leq \frac{1}{n} + 2n(\Delta - 1) \cdot \frac{\big(1 + 2n(\Delta - 1)\gamma\big)^2}{n(n-1)} \cdot \frac{1/n}{1 - \frac{1}{n(n-1)}}$$

and routine calculations show that this is at most $203/(128n)$.

For the lower bound, we use Corollary 7.9 to get:

$$P(g) \geq \mu(g) - \sum_{f \in \mathcal{F}} N(g, f \cap g)\kappa(f, \Omega \setminus g)$$

We will bound $N(g, f \cap g)$ via Theorem 7.6. Let $\mathcal{G} = \{f \in \mathcal{F} : f \cap g \neq \emptyset\}$. Consider some such flaw $f$ corresponding to atoms $f_1, f_2$. First, suppose $f_1, f_2, g$ are distinct. Then Corollary 7.7 gives

$$N(g, f \cap g) \leq \frac{1}{n(n-1)(n-2)} \big(1 + \sum_{f': f' \sim g} \Psi_{\mathcal{G}}(f')\big) \prod_{i=1}^{2} \big(1 + \sum_{f': f' \sim f_i} \Psi_{\mathcal{G}}(f')\big)$$

We can estimate $\sum_{f': f' \sim f_i} \Psi_{\mathcal{G}}(f') \leq \sum_{f': f' \sim f_i} \Psi_{\mathcal{F}}(f') \leq 2n(\Delta - 1)\gamma$ for each $i = 1, 2$. Also, $\mathcal{G}$ does not contain any flaws $f$ with $f \sim g$, so the term $\sum_{f': f' \sim g} \Psi_{\mathcal{G}}(f')$ contributes nothing. Overall we have $N(g, f \cap g) \leq \frac{(1+2n(\Delta-1)\gamma)^2}{n(n-1)(n-2)}$. There are at most $\frac{n^2(\Delta-1)}{2}$ flaws of this kind. By Proposition 7.10, each such flaw $f$ has $\kappa(f, \Omega \setminus g) \leq \frac{1 - \frac{(1/n(n-1))(1/n)}{1/n(n-1)(n-2)}}{1 - 1/n(n-1)} = \frac{2/n}{1 - 1/n(n-1)}$.

Next, suppose that say $f_1 = g$. Then, by the same reasoning as above, Corollary 7.7 gives $N(g, f \cap g) = N(g, f) \leq \frac{(1+2n(\Delta-1)\gamma)}{n(n-1)}$. There are at most $(\Delta - 1)$ flaws of this kind, and each trivially has $\kappa(f, \Omega \setminus g) \leq 1$.

Putting all terms together, we have

$$P(g) \geq 1/n - n^2(\Delta-1)/2 \cdot \frac{(1+2n(\Delta-1)\gamma)^2}{n(n-1)(n-2)} \cdot \frac{2/n}{1-\frac{1}{n(n-1)}} - (\Delta-1)\frac{(1+2n(\Delta-1)\gamma)}{n(n-1)} \cdot 1$$

which is easily seen to be at least $\frac{17}{32n}$. □

Note that Theorem 4.2 would yield a weaker bound $P(\pi x = y) \leq \frac{16}{9n}$, and Theorem 7.6 directly would yield a weaker bound $P(\pi x = y) \leq \frac{5}{3n}$. It is not known if a constant term better than 16/9 can be shown for the LLL-distribution.

# A Proof of Theorem 7.6

For brevity throughout, for a stable set $I$ of $\mathcal{A}$, we write $e_I$ as shorthand for $e_{\langle I \rangle}$.

As an easy warm-up exercise, we consider the variable setting. To recall and set notation, $\Omega$ is the Cartesian product distribution on tuples $X = (X_1, \ldots, X_n)$. For each index $i = 1, \ldots, n$ and value $y$, there is atomic event $X_i = y$; for brevity in this section, we denote this atom by $[i, y]$. Its resampling oracle draws $X_i$ again from its original distribution. We have $[i, y] \sim [i', y']$ if $i = i'$ and $y \neq y'$.

Proposition A.1. For any $\mathcal{A}$-stable multiset $I$, and corresponding set $\bar{I}$ (i. e., where we keep at most one copy of each element in $I$), there holds $A_I e_C \propto e_D$ for stable set $D = \bar{I} \cup (C \setminus \Gamma(\bar{I}))$.

Proof. We show this by induction on $I$. The base case $|I| = 0$ is trivial since then $D = C$. For the induction step, consider some atom $f = [i, y]$, and let $I' = I \uplus \{f\}$. By induction hypothesis, we have $A_{I'} = A_f A_I e_C \propto A_f e_D$ for $D = \bar{I} \cup (C \setminus \Gamma(\bar{I}))$. So it suffices to show that $A_f e_D \propto e_{D'}$ for $D' = \bar{I}' \cup (C \setminus \Gamma(\bar{I}')) = D \cup \{f\} \setminus \Gamma(f)$.

Consider a state $X$. If $X \notin f$, then $e_X^\top A_f e_D = 0 = e_X^\top e_{D'}$. Similarly, if $X \notin f'$ for $f' \in D \setminus \Gamma(f)$, then also $e_X^\top A_f e_D = 0 = e_X^\top e_{D'}$ since resampling $f$ cannot move the state to $f'$. Thus, we suppose that $X \in \langle D' \rangle$. Since $C$ is a stable set, it has at most one neighbor of $f$. If $C \cap \Gamma(f) = \emptyset$, then $e_X^\top A_f e_C = 1$ for any such state $X$; otherwise, if $C \cap \Gamma(f) = \{[i, y']\}$, then $e_X^\top A_f e_C = \Pr_\Omega(X_i = y')$ for any such state $X$. In either case, it is a scalar which is constant for all $X \in \langle D' \rangle$. □

Proof of Theorem 7.6 for the variable setting. Enumerate $I = \{f_1, \ldots, f_k\}$ to satisfy Proposition 7.5, where $f_i = \langle F_i \rangle$. For each $i$, let $J_i = F_1 \uplus \cdots \uplus F_i$ and let $\bar{J}_i = F_1 \cup \cdots \cup F_k$. We claim that, for each $i = 1, \ldots, k$, we have

$$\bar{J}_i \cap \Gamma(C) \neq \bar{J}_{i-1} \cap \Gamma(C) \tag{4}$$

For, suppose that $\bar{J}_i \cap \Gamma(C) = \bar{J}_{i-1} \cap \Gamma(C)$, and define $I' = \{f_1, \ldots, f_{i-1}\}$. By Theorem 6.7 we have $A_{I'} \propto A_{J_{i-1}}$ and by Proposition A.3, we have $A_{J_{i-1}} e_E \propto e_{D'}$ where $D' = \bar{J}_{i-1} \cup (C \setminus \Gamma(\bar{J}_{i-1})$. Thus, we can write $A_{I'} e_E = p e_{D'}$ for some scalar $p \geq 0$. Now consider any state $X$; we claim that

$$e_X^\top A_{f_i} A_{I'} \leq p \cdot e_X^\top e_{D'} \tag{5}$$

Let $D = \bar{J}_i \cup (C \setminus \Gamma(\bar{J}_i))$; since $\bar{J}_i \cap \Gamma(C) = \bar{J}_{i-1} \cap \Gamma(C)$ we have $D \supseteq D'$. So the LHS of Eq. Eq. (5) is zero unless $X \in \langle D \rangle \subseteq \langle D' \rangle$. On the other hand, for $X \in \langle D \rangle$, the substochasticity of matrix $A_{f_i}$ gives $e_X^\top A_{f_i} A_{I'} e_E = e_X^\top A_{f_i} \cdot p e_{D'} \leq p$ as desired.

So, if $\bar{J}_i \cap \Gamma(C) = \bar{J}_{i-1} \cap \Gamma(C)$, we would have $A_{f_i} A_{I'} \preceq A_{I'}$, contradicting Proposition 7.5. Thus, contrariwise, we have established Eq. (4) for $i = 1, \ldots, k$. We define the function $\phi$ by setting $\phi(f_i) = g_i$ where $g_i$ is an arbitrary element in $(\bar{J}_i \cap \Gamma(C)) \setminus (\bar{J}_{i-1} \cap \Gamma(C))$. □

We now turn to the much harder permutation setting. To recall and set notation, $\Omega$ is the uniform distribution on permutations $\pi$ on $\{1, \ldots, n\}$. For each pair $(x, y)$, there is atomic event $\pi x = y$; for brevity in this section, we denote this event by $[x, y]$. Its resampling oracle sets $\pi \leftarrow (y\ z)\pi$, for $z$ drawn uniformly from $\{1, \ldots, n\}$. We have $[x, y] \sim [x', y']$ if exactly one of the following holds: (i) $x = x'$ or (ii) $y = y'$.

Proposition A.2. Let $f = [x, y]$ be an atom of $\mathcal{A}$ and let $C$ be a stable set of $\mathcal{A}$. Then $A_f e_C \propto e_{C'}$ for the $\mathcal{A}$-stable set $C'$ obtained from $C$ as follows.

- If $C$ contains two neighbors $f_1 = [x, y_1], f_2 = [x_2, y]$ of $f$, then $C' = C \cup \{[x_2, y_1], f\} \setminus \Gamma(f)$.
- Otherwise, $C' = C \cup \{f\} \setminus \Gamma(f)$.

Proof. We want to show that there is a scalar $p$ with $e_\pi^\top A_f e_C = p \cdot e_\pi e_{C'}$ for all states $\pi$. If $\pi \notin f$, then $e_\pi^\top A_f e_C = 0 = e_\pi^\top e_{C'}$. Similarly, if $\pi \notin f'$ for $f' \in C \setminus \Gamma(f)$, then also $e_\pi^\top A_f e_C = 0 = e_\pi^\top e_{C'}$ since resampling $f$ cannot move the state to $f'$. Thus, we suppose that $\pi \in \langle C \cup \{f\} \setminus \Gamma(f) \rangle$.

Now suppose we resample $f$ to $\pi' = (y\ z)\pi$. We consider the cases in turn:

- If $C$ contains two neighbors $f_1 = [x, y_1], f_2 = [x_2, y]$, then we claim that $\pi' \in \langle C \rangle$ precisely when $z = \pi x_2 = y_1$. For, in order to get $\pi' \in f_1$, we must have $\pi' x = y_1$. Since $\pi x = y$, this implies $(y\ z)y = y_1$, i. e., $z = y_1$. Thus, $\pi' = (y\ y_1)\pi$. To get $\pi' \in f_2$, we need $y = \pi' x_2 = (y\ y_1)\pi x_2$, i. e., $\pi x_2 = y_1$. In particular, we must have $\pi \in [x_2, y_1]$, and for any such $\pi$ we have $e_\pi^\top A_f e_C = 1/n$ and $e_\pi^\top e_{C'} = 1$.

- If $C$ has a single neighbor $f_1 = [x, y_1]$ of $f$, then $\pi' \in f_1$ precisely if $y_1 = z$. Similarly, if $C$ has a single neighbor $f_2 = [x_2, y]$ of $f$, then $\pi' \in f_2$ precisely if $z = \pi x_2$. In either case, we have $e_\pi^\top A_f e_C = 1/n$ for all such $\pi$ and also $e_\pi^\top e_{C'} = 1$.

- If $f \in C$, then $\pi'$ is in $\langle C \rangle$ iff $z = y$ and $\pi' = \pi \in \langle C \rangle$. Then $e_\pi^\top A_f e_C = \frac{1}{n} e_C$. Note that, because $C$ is a stable set, this case implies that $C$ has no neighbors of $f$.

- If $C \cap \overline{\Gamma}(f) = \emptyset$, then $\pi'$ is in $\langle C \rangle$ iff $z \notin \{y_1, \ldots, y_k\}$ where $C = \{[x_1, y_1], \ldots, [x_k, y_k]\}$. Thus $e_\pi^\top A_f e_C = \frac{n-k}{n}$ and $e_\pi^\top e_{C'} = 1$. □

Proposition A.3. Given stable multisets $C, I$ of $\mathcal{A}$, let $\bar{I}$ denote the corresponding set of $I$ (i. e., with at most one copy of each element). Define a bipartite graph $G_I^C$ with left-vertex-set $C$ and right-vertex-set $\bar{I}$, with an edge on $f, f'$ iff $f \sim f'$, and let $\tau^C(\bar{I})$ be the size of a maximum matching in $G_I^C$.

For such multisets $I, C$, there is an associated stable set $D_I^C$ of $\mathcal{A}$ with $A_I e_C \propto e_{D_I^C}$. Furthermore, if $\tau^C(\bar{I} \cup \{f\}) = \tau^C(\bar{I})$, then $D_{I \uplus \{f\}}^C = D_I^C \cup \{f\}$.

Proof. Throughout this proof, we fix $C$ and write $D_I, \tau(\bar{I}), G_I$ instead of $D_I^C, \tau^C(\bar{I}), G_I^C$ etc.

Since $C$ and $\bar{I}$ are stable sets, the graph $G_I$ has degree at most two — each node $[x, y]$ has at most one neighbor of the form $[x', y]$ and at most one neighbor of the form $[x, y']$. So $G_I$ decomposes into paths and cycles and any maximal path of $G_I$ which starts and ends at left-nodes (which we call a $C$-path), can be written uniquely as

$$[x_1, y_1], [x_1, y_2], [x_2, y_2], \ldots, [x_k, y_{k-1}], [x_k, y_k];$$

We define $H_I$ to be the set of atoms $[x_k, y_1]$ for each such $C$-path; this includes the case $k = 1$ where $[x_1, y_1]$ is an isolated $C$-node. We define $D_I = \bar{I} \cup H_I$; note that $D_I$ is an $\mathcal{A}$-stable set.

We first show that $A_I e_C \propto e_{D_I}$ by induction on $|I|$. The base case $I = \emptyset$ holds since then $D_I = C$. For the induction step, let $I' = I \uplus \{f\}$.

By induction hypothesis, we have $A_{I'}e_C = A_f A_I e_C \propto A_f e_U$ for $U = D_I$. This in turn is proportional to $e_{U'}$ for the stable set $U'$ obtained from $U, f$ according to the rules given in Proposition A.2. There are three cases.

If $U$ has no neighbors of $f$, or if $f \in D_I$, then $U' = U \cup \{f\}$, and $H_{I'} = H_I \cup \{f\}$. So indeed $U' = D_{I'} = D_I \cup \{f\}$. Thus, we assume for the remainder that $f \notin I$.

If $U$ has one neighbor $g = [x_1, y_1]$ of $f$, then since $\bar{I}$ is a stable set, it must have $g \notin \bar{I}$, i. e., $G_I$ contains a $C$-path $P$ with endpoints $x_1, y_1$. In $G_{I'}$, this $C$-path now terminates in a degree-one left-node $[x, y]$, and $P$ is no longer a $C$-path of $G_{I'}$. So $D_I \cup \{f\} \setminus \{g\} = U' = D_{I'}$.

If $U$ has two neighbors $g_1 = [x, y_1], g_2 = [x_2, y]$, then again since $\bar{I}$ is a stable set these must correspond to $C$-paths in $G_I$. Thus, $G_I$ has two $C$-paths with endpoints $x, y_1$ and $x_2, y$ respectively. Now $G_{I'}$ has a new left-node $[x, y]$. This merges the two $C$-paths into a single new $C$-path with endpoints $x_2, y_1$. Thus again $D_{I'} = D_I \cup \{f, [x_2, y_1]\} \setminus \{g_1, g_2\} = U'$.

This completes the induction. Next, suppose $\tau^C(\bar{I} \cup \{f\}) = \tau^C(\bar{I})$. If $f \in C$ or $f \in I$, then $f \in D_I$ and we have already seen that $D_{I'} = D_I = D_I \cup \{f\}$. So suppose $f$ is not in $C$ or $I$, and let $\tilde{G}'$ denote the connected component of $G_{I'}$ containing $f$, and let $\tilde{G}$ be the graph obtained by deleting $f$ from $\tilde{G}'$. Since $\tilde{G}'$ is a path or cycle, and $\tilde{G}$ is obtained by deleting a right-node, the only way that $\tilde{G}$ and $\tilde{G}'$ can have the same matching size is if $\tilde{G}'$ is a path starting and ending at right-nodes. In this case, neither $\tilde{G}'$ nor $\tilde{G}'$ have any $C$-paths. Hence we get $H_{I \uplus \{f\}} = H_I$. □

Proof of Theorem 7.6 for the permutation setting. Enumerate $I = \{f_1, \ldots, f_k\}$ to satisfy Proposition 7.5, where $f_i = \langle F_i \rangle$. For each $i$, let $J_i = F_1 \uplus \cdots \uplus F_i$ and let $\bar{J}_i = F_1 \cup \cdots \cup F_k$. We claim that, for each $i = 1, \ldots, k$, we have

$$\tau^C(\bar{J}_i) > \tau^C(\bar{J}_{i-1}) \tag{6}$$

For, suppose that $\tau^C(\bar{J}_i) = \tau^C(\bar{J}_{i-1})$, and define $I' = \{f_1, \ldots, f_{i-1}\}$. We have $A_{I'} \propto A_{J_{i-1}}$ and by Proposition A.3, we have $A_{J_{i-1}} e_E \propto e_{D'}$ for some stable set $D'$ of $\mathcal{A}$. Thus, we can write $A_{I'} e_E = p e_{D'}$ for some scalar $p \geq 0$. Now consider any state $\pi$; we claim that

$$e_\pi^\top A_{f_i} A_{I'} \leq p \cdot e_\pi^\top e_{D'} \tag{7}$$

Again, by Proposition A.3, we have $A_{f_i} A_{I'} \propto e_D$ for a stable set $D$; since $\tau^C(F_i \cup \bar{J}_{i-1}) = \tau^C(\bar{J}_{i-1})$ we have $D = D' \cup F_i$. So the LHS of (7) is zero unless $\pi \in \langle D \rangle \subseteq \langle D' \rangle$. On the other hand, for $\pi \in \langle D \rangle$, the substochasticity of matrix $A_{f_i}$ gives $e_\pi^\top A_{f_i} A_{I'} e_E = e_\pi^\top A_{f_i} \cdot p e_{D'} \leq p$ as desired.

So, if $\tau^C(\bar{J}_i) = \tau^C(\bar{J}_{i-1})$, we would have $A_{f_i} A_{I'} \preceq A_{I'}$, contradicting Proposition 7.5. Thus, contrariwise, we have established Eq. (6) for $i = 1, \ldots, k$. So, for each $i$ there is $g_i \in F_i$ and $F_i' \subseteq F_i \setminus \{g_i\}$ with $\tau^C(\bar{J}_{i-1} \cup F_i' \cup \{g_i\}) > \tau^C(\bar{J}_{i-1} \cup F_i')$.

It is known (see, e. g., [25, Example 1.4]) that $\tau^C$ is a submodular set function for fixed $C$. Hence,

$$1 = \tau^C(\bar{J}_{i-1} \cup F_i' \cup \{g_i\}) - \tau^C(\bar{J}_{i-1} \cup F_i') \leq \tau^C(\{g_1, \ldots, g_{i-1}\} \cup \{g_i\}) - \tau^C(\{g_1, \ldots, g_{i-1}\})$$

since $\{g_1, \ldots, g_{i-1}\} \subseteq \bar{J}_{i-1}$. So $\tau^C(\{g_1, \ldots, g_k\}) = k$ and $G^C_{\{g_1, \ldots, g_k\}}$ has a matching $M$ of size $k$. We define the function $\phi$ by setting $\phi(f_i) = c_i$ where $g_i$ is matched to $c_i$ in $M$. □

The clique-perfect-matching setting is very similar to the permutation setting. Here, $\Omega$ is the uniform distribution on perfect matchings $M$ of the $n$-clique. For each pair $(x, y)$ with $x \neq y$, there is an atom that $M \supseteq \{x, y\}$; we denote this by $[x, y]$. Note that $[x, y] = [y, x]$. For $x < y$, the resampling oracle is to draw $z$ uniformly from $\{1, \ldots, n\} \setminus \{x\}$ and set $M \leftarrow (y\ z)M$. We define $[x, y] \sim [x', y']$ iff $|\{x, y\} \cap \{x', y'\}| = 1$.

Proposition A.4. Let $f = [x, y]$ be an atom of $\mathcal{A}$ where $x < y$ and let $C$ be a stable set of $\mathcal{A}$. Then $A_f e_C \propto e_{C'}$ for the stable set $C'$ obtained from $C$ as follows.

- If $C$ contains exactly two neighbors $f_1 = [x, y_1], f_2 = [x_2, y]$ of $f$, then $C' = C \cup \{f, [x_2, y_1]\} \setminus \Gamma(f)$.
- Otherwise, $C' = C \cup \{f\} \setminus \Gamma(f)$.

Proposition A.5. Given stable multisets $C, I$ of $\mathcal{A}$, let $\bar{I}$ denote the corresponding set of $I$. Define a bipartite graph $G^C_I$ with left-vertex-set $C$ and right-vertex-set $\bar{I}$, with an edge on $f, f'$ iff $f \sim f'$, and let $\tau^C(\bar{I})$ be the size of a maximum matching in $G^C_I$.

For such multisets $I, C$, there is an associated stable set $D^C_I$ of $\mathcal{A}$ with $A_I e_C \propto e_{D^C_I}$. Furthermore, if $\tau^C(\bar{I} \cup \{f\}) = \tau^C(\bar{I})$, then $D^C_{I \uplus \{f\}} = D^C_I \cup \{f\}$.

We omit the proofs of Proposition A.4 and Proposition A.5, as well as the remainder of the proof of Theorem 7.6, as they are precisely analogous to the permutation setting.

# References


[1] Dimitris Achlioptas and Fotis Iliopoulos: Random walks that find perfect objects and the Lovász Local Lemma. J. ACM, 63(22):1–29, 2016. Preliminary version in FOCS'14. [doi:10.1145/2818352]

[2] Dimitris Achlioptas, Fotis Iliopoulos, and Vladimir Kolmogorov: A local lemma for focused stochastic algorithms. SIAM J. Comput., 48(5):1583–1602, 2019. [doi:10.1137/16M109332X]

[3] Dimitris Achlioptas, Fotis Iliopoulos, and Alistair Sinclair: Beyond the Lovász Local Lemma: Point to set correlations and their algorithmic applications. In Proc. 60th FOCS, pp. 725–744. IEEE Comp. Soc., 2019. [doi:10.1109/FOCS.2019.00049]

[4] Noga Alon, Joel Spencer, and Prasad Tetali: Covering with Latin transversals. Discr. Appl. Math., 57(1):1–10, 1995. [doi:10.1016/0166-218X(93)E0136-M]

[5] Rodrigo Bissacot, Roberto Fernández, Aldo Procacci, and Benedetto Scoppola: An improvement of the Lovász Local Lemma via cluster expansion. Combin. Probab. Comput., 20(5):709–719, 2011. [doi:10.1017/S0963548311000253]

[6] Karthekeyan Chandrasekaran, Navin Goyal, and Bernhard Haeupler: Deterministic algorithms for the Lovász Local Lemma. SIAM J. Comput., 42(6):2132–2155, 2013. Preliminary version in SODA'10. [doi:10.1137/100799642]

[7] Antares Chen, David G. Harris, and Aravind Srinivasan: Partial resampling to approximate covering integer programs. Random Struct. Algor., pp. 68–93, 2021. Preliminary version in SODA'16. [doi:10.1002/rsa.20964]

[8] Kai-Min Chung, Seth Pettie, and Hsin-Hao Su: Distributed algorithms for the Lovász Local Lemma and graph coloring. Distrib. Comput., 30(4):261–280, 2017. Preliminary version in PODC'14. [doi:10.1007/s00446-016-0287-6]

[9] Paul Erdős and László Lovász: Problems and results on 3-chromatic hypergraphs and some related questions. In András Hajnal, Richard Rado, and Vera T. Sós, editors, Infinite and finite sets: To Paul Erdős

on his 60th birthday, volume II of Colloq. Math. Soc. János Bolyai, 10, pp. 609–627. János Bolyai Math. Soc. and North-Holland, 1975.

[10] Paul Erdős and Joel Spencer: Lopsided Lovász Local Lemma and Latin transversals. Discr. Appl. Math., 30(2–3):151–154, 1991. [doi:10.1016/0166-218X(91)90040-4]

[11] Bernhard Haeupler and David G. Harris: Parallel algorithms and concentration bounds for the Lovász Local Lemma via witness DAGs. ACM Trans. Algorithms, 13(4):53:1–25, 2017. Preliminary version in SODA'17. [doi:10.1145/3147211]

[12] Bernhard Haeupler, Barna Saha, and Aravind Srinivasan: New constructive aspects of the Lovász Local Lemma. J. ACM, 58(6):28:1–28, 2011. Preliminary version in FOCS'10. [doi:10.1145/2049697.2049702]

[13] David G. Harris: Lopsidependency in the Moser–Tardos framework: Beyond the lopsided Lovász Local Lemma. ACM Trans. Algorithms, 13(1):17:1–26, 2016. Preliminary version in SODA'15. [doi:10.1145/3015762]

[14] David G. Harris: New bounds for the Moser–Tardos distribution. Random Struct. Algor., 57(1):97–131, 2020. [doi:10.1002/rsa.20914]

[15] David G. Harris: Oblivious resampling oracles and parallel algorithms for the Lopsided Lovász Local Lemma. ACM Trans. Algorithms, 17(1):1:1–32, 2021. Preliminary version in SODA'19. [doi:10.1145/3392035]

[16] David G. Harris, Fotis Iliopoulos, and Vladimir Kolmogorov: A new notion of commutativity for the algorithmic Lovász Local Lemma. In Proc. 25th Internat. Conf. on Randomization and Computation (RANDOM'21), pp. 31:1–25. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2021. [doi:10.4230/LIPIcs.APPROX/RANDOM.2021.31]

[17] David G. Harris and Aravind Srinivasan: Algorithmic and enumerative aspects of the Moser–Tardos distribution. ACM Trans. Algorithms, 13(3):33:1–40, 2017. Preliminary version in SODA'16. [doi:10.1145/3039869]

[18] David G. Harris and Aravind Srinivasan: A constructive Lovász Local Lemma for permutations. Theory of Computing, 13(17):1–41, 2017. Preliminary version in SODA'14. [doi:10.4086/toc.2017.v013a017]

[19] David G. Harris and Aravind Srinivasan: The Moser–Tardos framework with partial resampling. J. ACM, 66(5):36:1–45, 2019. Preliminary version in FOCS'13. [doi:10.1145/3342222]

[20] Nicholas J. A. Harvey and Jan Vondrák: An algorithmic proof of the Lovász Local Lemma via resampling oracles. SIAM J. Comput., 49(2):394–428, 2020. Preliminary version in FOCS'15. [doi:10.1137/18M1167176]

[21] Fotis Iliopoulos: Commutative algorithms approximate the LLL-distribution. In Proc. 22nd Internat. Conf. on Randomization and Computation (RANDOM'18), pp. 44:1–20. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2018. [doi:10.4230/LIPIcs.APPROX-RANDOM.2018.44]

[22] Fotis Iliopoulos and Alistair Sinclair: Efficiently list-edge coloring multigraphs asymptotically optimally. Random Struct. Algor., 61(4):724–753, 2022. Preliminary version in SODA'20. [doi:10.1002/rsa.21074]

[23] Kashyap Babu Rao Kolipaka and Mario Szegedy: Moser and Tardos meet Lovász. In Proc. 43rd STOC, pp. 235–244. ACM Press, 2011. [doi:10.1145/1993636.1993669]

[24] Vladimir Kolmogorov: Commutativity in the algorithmic Lovász Local Lemma. SIAM J. Comput., 47(6):2029–2056, 2018. Preliminary version in FOCS'16. [doi:10.1137/16M1093306]

[25] László Lovász: Submodular functions and convexity. In Achim Bachem, Bernhard Korte, and Martin Grötschel, editors, Mathematical Programming: The State of the Art, pp. 235–257. Springer, 1983. [doi:10.1007/978-3-642-68874-4_10]

[26] Robin A. Moser and Gábor Tardos: A constructive proof of the general Lovász Local Lemma. J. ACM, 57(2):11:1–15, 2010. Preliminary version in STOC'09. [doi:10.1145/1667053.1667060]

[27] Wesley Pegden: An extension of the Moser–Tardos algorithmic local lemma. SIAM J. Discr. Math., 28(2):911–917, 2014. [doi:10.1137/110828290]